An inventory-state exact algorithm for bicycle redistribution with ratio-Gini and target-deviation costs

26 October 2026

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Abstract

Static bicycle redistribution couples final station inventories with vehicle routes, prefix loads and handling time. We minimize ratio-Gini dispersion plus weighted relative target deviation under empty departure, loaded return and unique nonzero one-way service. Our exact method shares finite inventory states between service and objective representations, uses problem-specific supply and support inequalities, and maintains complete Gini-interval coverage around native LP and MIP solves. We prove implicit service integrality and establish the physical embeddings and coverage conditions needed for certification. In historical twelve-input/two-Seed validation, the method obtains 17/24 qualified certificates versus 9/24 for a cold compact control. Final implementation confirmation on six known roles obtains 12/14 versus 9/14, while retaining two substantive paid-start compact reversals and unresolved positive-objective 35-station endpoints. Six certified within-instance weight points show an equity-target trade-off on one input and a stable operating platform on another. The results support the integrated exact method within the specified model and test scope, and a bounded state-penalty study distinguishes relaxation improvement from complete-search performance.

1. Introduction

Redistribution vehicles can improve station availability only when the chosen inventory vector can be realized by feasible routes and loading operations. Supply must be collected before it is delivered, capacity applies at every route prefix, and handling consumes the same time budget as travel. When equity is measured by inventory-to-target ratios, the final inventory vector also determines both the numerator and denominator of the Gini coefficient. These dependencies make routing and inventory allocation inseparable.

We study static bicycle redistribution with empty vehicle departure, loaded return, and at most one nonzero one-way service at each station. The objective combines ratio-Gini dispersion and weighted relative target deviation. The model and HGA-TGBC framework originate in the direct same-problem project manuscript (*Equity-Aware Bike Repositioning in Station-Based Bike-Sharing Systems* 2026). The present contribution is an exact solution method for that operational setting. The supplied decoder remains a feasible-solution component; the exact method supplies the bounds, coverage and termination arguments needed to distinguish a good plan from a certified optimum.

Our first contribution is a shared finite-state representation of final inventory, service and objective information. Unique service and binary direction imply integer pickup and delivery even when these quantities are declared continuous. Linking the same states to movement and visits prevents the relaxation from treating service decisions

independently of the inventory mixtures used in the objective. We develop this representation with ordered supply compatibility and support/handling inequalities, and state their assumptions and physical embeddings.

Second, the Gini-interval tailored branch-and-cut method (GI-TBC, implemented as S2) combines a fully paid feasible startup, the shared representation, valid inequalities and complete interval coverage. The controller preserves the physical problem while refining objective intervals. Its correctness depends on applicable bounds and complete coverage; numerical source qualification and physical solution verification are stated separately.

Third, the computational study measures complete-method cost against cold compact and independently paid-start compact controls. Historical breadth and final implementation confirmation retain their own versions and denominators. The study identifies substantial certification and runtime advantages, paid-start reversals and a positive-objective size boundary. A within-instance weight experiment relates the equity-target trade-off to actual vehicle operations. A bounded formulation study explains why retaining more state information can improve an LP without guaranteeing a faster complete algorithm.

1.1 Relation to previous work

Static bike redistribution has been studied under several target, service and vehicle conventions. Raviv et al. develop static repositioning models; Chemla et al. study static rebalancing; Erdogan et al. address demand intervals and an exact rebalancing algorithm (Raviv et al. 2013; Chemla et al. 2013; Erdoğan et al. 2014, 2015). Schuijbroek et al. connect inventory requirements to routing (Schuijbroek et al. 2017). Forbidden temporary operations, repeated visits, robust inventory requirements and demand-interval disjunctions give related but different feasible sets (Bruck et al. 2019; Bulhões et al. 2018; Bruck et al. 2025; Ikonen et al. 2025). The HGA-TGBC manuscript is the direct same-problem antecedent; the other studies supply routing and formulation context rather than matched performance baselines.

Station-ratio dispersion measures a different object from vehicle-workload equity (Matl et al. 2018). Recent fairness-aware rebalancing and equity-based micromobility models motivate distributional objectives, while their distinct inventory, vehicle and operational definitions preclude a claim of model equivalence based on their abstracts alone (Qiao et al. 2026; Caggiani et al. 2026). Here zero Gini means equal proportional inventory, which can still represent inadequate stock. The deviation term gives a separate incentive to approach station targets.

McCormick envelopes, value disjunctions and convex state epigraphs are established formulation tools (McCormick 1976; Gupte et al. 2013; Köppe et al. 2008; Boyd and Vandenberghe 2004). Our claim concerns their connection to this problem's service structure and the resulting complete method. The state-penalty comparison in Section 6 applies Jensen's inequality and an elementary projection construction; it is not a new general theory of aggregation. Source access and the remaining narrowly defined bibliographic questions are documented in the supplement.

2. Physical model and objective

Let N contain n stations and let 0 be the depot. Station i has integer initial inventory b_i, capacity C_i, integer target $D_i > 0$, and weight $\omega_i \geq 0$. Vehicle k has capacity Q_k and route-duration limit T. Routes depart empty. A station can be served by at most one vehicle, in one direction, with a nonzero operation. Loaded return is allowed and its unloading time is counted. The depot enters neither target deviation nor dispersion.

With pickup p_{ki} and delivery d_{ki},

$$Y_i = b_i + \sum_k (d_{ki} - p_{ki}), \qquad R_k = \sum_i (p_{ki} - d_{ki}) \in [0, Q_k], \qquad \sum_i Y_i = \sum_i b_i - \sum_k R_k.$$

Thus total final station inventory is bounded above by $B = \sum_i b_i$, rather than fixed at B. Vehicle capacity limits the instantaneous prefix load, rather than cumulative pickup. If pickup and unloading/delivery take t_p, t_d per unit, complete handling time is

$$t_p \sum_i p_{ki} + t_d \left(\sum_i d_{ki} + R_k \right) = (t_p + t_d) \sum_i p_{ki}.$$

Travel plus this quantity cannot exceed T. The tested distance convention is Euclidean travel time from serialized metre coordinates at 1.5 m/s. Inventory, route, assignment, direction and prefix-load logic retain their original integer domains. Continuous pickup/delivery variables preserve the integer physical operations because unique one-way service and integer Y_i fix the sole inventory difference. Appendix A gives the complete formulation and embedding proof.

Define

$$r_i = Y_i/D_i, \quad S = \sum_i r_i, \quad A = \sum_{i<j} |r_i - r_j|, \quad P = \sum_i \omega_i |r_i - 1|,$$

$$G = \begin{cases} A/(nS), & S > 0, \\ 0, & S = 0, \end{cases} \qquad F = G + \lambda P, \quad \lambda \geq 0.$$

This is ratio dispersion and two-sided relative target deviation. It is not a direct measure of population equity, rental failure or travel burden. Put $\Omega = \sum_i \omega_i > 0$, $P_{\text{avg}} = P/\Omega$ and $\rho = \lambda\Omega$. Then $F = G + \rho P_{\text{avg}}$. The generator normalizes weights by their maximum, so equal λ across inputs does not hold ρ fixed. Sensitivity in Section 5.4 changes λ within a fixed physical instance.

3. Shared inventory states and the S2 algorithm

3.1 Discrete states and implicit service integrality

For the legal finite inventory domain $\mathcal{Y}_i$, binary selectors s_{iy} and continuous q_{iy} represent stock and products on a Gini interval $[\alpha, \beta]$:

$$\sum_y s_{iy} = 1, \quad Y_i = \sum_y y s_{iy}, \quad \alpha s_{iy} \leq q_{iy} \leq \beta s_{iy}, \quad \sum_y q_{iy} = g, \quad Z_i = \sum_y y q_{iy}.$$

At an integer state, $Z_i = gY_i$. The objective epigraph uses

$$H_{ij} \geq r_i - r_j, \quad H_{ij} \geq r_j - r_i, \qquad e_i \geq r_i - 1, \quad e_i \geq 1 - r_i,$$
$$\sum_{i<j} H_{ij} \leq n \sum_i Z_i/D_i, \qquad \min f = g + \lambda \sum_i \omega_i e_i.$$

For $S > 0$, integer state selection makes the numerator inequality imply $g \geq G_{\text{true}}$, so every integer completion satisfies $f \geq F_{\text{true}}$. Conversely a physical fleet assigned to its true-G interval has an objective-preserving extension: select its final inventories, set $g = G_{\text{true}}$, set each selected $q_{iy} = g$, and use the true absolute differences and deviations. At $S = 0$, all ratios are zero and $g = 0$ supplies the physical extension. Valid domain and strengthening rows retain that extension, as established in Appendix A.

Let z_{ki} be the binary service assignment. The same state supports the service equations

$$\sum_k p_{ki} = \sum_y (b_i - y)^+ s_{iy}, \quad \sum_k d_{ki} = \sum_y (y - b_i)^+ s_{iy}, \quad \sum_k z_{ki} = \sum_{y \neq b_i} s_{iy}.$$

These links connect objective mixtures to movement and service. They are not the convex hull of all vehicle routes. In isolation, a finite-state product block with the same extreme stock bounds projects to the McCormick rectangle: its four corner graph points give one inclusion, and every state line lies in the rectangle hull, giving the other. An advantage must therefore come from the links, domain information or complete search, rather than the VD-P label alone.

Lemma 1 (implicit service integrality). Suppose initial and final inventories are integer, a station is assigned to at most one vehicle, and a binary direction permits either pickup or delivery at its unique service. Continuous pickup and delivery variables then take integer values at every integer solution of the remaining model.

Proof. An unassigned vehicle has zero quantities. At the unique assigned vehicle, pickup direction fixes delivery to zero, so inventory balance gives $p_{ki} = b_i - Y_i$. Delivery direction fixes pickup to zero and gives $d_{ki} = Y_i - b_i$. Each is integer. Integer load variables are unnecessary for this implication. The claim concerns integer feasible solutions; it does not equate the continuous relaxations of alternative formulations. □

For an integer service, total movement is $|b_i - Y_i|$, and nonzero one-way service occurs exactly when $Y_i \neq b_i$. Hence the shared equations above reproduce the original operations without changing the physical problem. They also couple fractional inventory mixtures to the movement and visits that must support them.

3.2 Ordered supply and support/handling information

Let τ be metric travel time and $h = t_p + t_d$. For each receiving station j, define

$$\mathcal{C}_{kj} = \{i \neq j : \tau_{0i} + \tau_{ij} + \tau_{j0} + h \leq T\}.$$

Proposition 1 (ordered supply compatibility). Every physical fleet with empty departure satisfies

$$d_{kj} \leq \sum_{i \in \mathcal{C}_{kj}} p_{ki}.$$

Proof. Label the units collected by a vehicle. In route order, match each delivered unit to an earlier collected unit that has not yet been delivered. Nonnegative prefix loads guarantee such a match. A unit delivered at j from source i entails the ordered subsequence $(0, i, j, 0)$, whose travel lower bound follows by metric shortcutting. Collecting at least that unit incurs handling of at least h, including delivery or final unloading. Thus its source lies in $\mathcal{C}_{kj}$. Counting distinct matched units proves the inequality. Unmatched collected units can return to the depot. □

For a set W of served stations, let $\tau(W)$ be the shortest depot tour through W, computed by explicit orders for pairs or triples. Write $P_k^{\text{pick}} = \sum_i p_{ki}$. Nonzero integer service and empty departure give

$$|W| \leq \sum_{i \in W} (p_{ki} + d_{ki}) \leq 2P_k^{\text{pick}}, \qquad P_k^{\text{pick}} \geq \lceil |W|/2 \rceil.$$

Consequently $\tau(W) + h\lceil |W|/2 \rceil > T$ rules out serving all of W with that vehicle. Otherwise a conditional duration inequality combines the tour lower bound and pickup handling. Supply for W may come from outside W; cumulative pickup is not bounded by Q_k. Appendix A gives the exact conditional coefficients, deactivation proof and finite selection rule. S2 keeps pair support uniformly and omits triple support. These rows strengthen necessary physical conditions without replacing route-prefix capacity.

3.3 Complete startup and interval search

The startup H is a finite construction and descent procedure. It completes a joint insertion and 24 shuffled station/separator orders plus a constructive order, fully decodes them, applies guided first-improvement descent, and closes physical quantity/insertion and neutral balanced/exchange moves. The active path has no genetic generations. It provides a verified feasible upper bound and a possible MIP Start; it is never credited with a positive lower bound. Each S2 and paid-start compact run pays for its own complete H.

The controller starts from the complete potentially improving domain $[0, \min\{U, (n-1)/n\}]$. It maintains incumbent epoch, valid cutoff complements and complete interval coverage. A local bound is used only for its model, interval, cutoff and qualified source. Split children replace a parent atomically only after complete coverage and both results are checked. NEXT and child-target values are requests, rather than proof that a milestone was reached.

```
Algorithm 1  S2 / GI-TBC
1 Complete the run's own H; physically verify its fleet and upper bound U.
2 Initialize the complete potentially improving Gini interval, allowing
  the degenerate interval [0,0] when the own upper bound is zero.
3 While unresolved coverage and the common deadline permit:
    select the controlling leaf by the fixed bound/plateau rule;
    run its prescribed LP, lookahead/target or terminal MIP;
    verify any own fleet improvement and qualify the returned source;
    install valid scoped bounds or complete covering children;
    stop under the declared complete-proof or finite termination rules.
4 On source rejection, discard dependent closures and restore the unresolved
  full physical domain with its independent nonnegative floor and own U.
  A separately established zero proof can still finalize the original problem.
5 Release native resources and finalize the selected own fleet's physical check;
  apply the independent exact-zero rule, complete required output, and return
  certificate type, U/L and unresolved coverage.
```

Proposition 2 (coverage). In ideal arithmetic, a controller that preserves complete improving-domain coverage, retains valid bound/cutoff complements and closes a region only by an applicable proof provides a valid global lower bound. If it closes all relevant regions against a verified U, it proves optimality to its declared certificate margin.

Proof. Every improving physical solution belongs to the root domain because $F \geq G$ and $G \leq (n-1)/n$. A valid interval restriction and cutoff cover its regional solutions; the cutoff complement has floor equal to that cutoff. A complete split preserves the union. Taking the minimum over the valid regional floors and complements therefore bounds every physical solution from below. Closure against U eliminates every possible improvement to the margin. Revoking a source requires removing every conclusion that depends on it; an independent full-domain bound remains valid. □

This proposition does not turn floating-point native calculations into exact rational proofs. Numerical qualification, physical objective agreement and lifecycle evidence are additional implementation conditions. The final implementation separates ordinary positive-objective native/outer certificates from the independent zero rule. Supporting material records numerical anomalies and version corrections without making them algorithmic contributions.

3.4 Numerical qualification and termination

Independent zero termination. Under the stated nonnegative objective premises, a complete physically verified fleet with exact objective zero is globally optimal, independently of the native search source.

Proof. For every original physical solution, $F \geq 0$. The verified fleet gives $F^* \leq 0$, so $0 \leq F^* \leq 0$. With positive λ and positive weights, integer equality $Y_i = D_i$ for every station is a sufficient exact-zero condition. More generally equal exact ratios give zero Gini, and every positive-weight penalized station must meet its target. The implementation checks these conditions using integer stock arithmetic, then repeats the complete physical-fleet verification. A merely small positive floating objective does not establish exact zero. □

The numerical ε-certificate policy is separate. A source-rejected positive solution without a sufficient independent bound remains open. Certification cost includes the verification needed to establish the proof, normal native cleanup and required output. A later offline audit cannot backdate a production proof.

4. Computational design

P starts the original compact model independently with no startup or MIP Start. PS pays its own complete startup H and then solves that compact model. S2 pays its own H and full interval method. Each comparison uses single-threaded native solves, serial independent launches and a fixed parameter policy. No fleet, bound, tree or Start crosses methods, Seeds or objective weights. The complete production clock includes startup, modeling, online verification, solving, proof formation, cleanup and required output; subsequent offline verification has its own measured clock.

Table 1 identifies the evidence versions. Solver Seeds repeat a fixed problem and are not independent problem samples. The R134 confirmation targets six known roles, and the AP study targets structural and protection roles; neither is described as a random holdout.

Table 1. Evidence identity and purpose. Historical outcomes retain their original admission policies.

Version	Algorithm or change	Evidence used here
R132	Frozen S2 and compact controls	Twelve inputs at two Seeds; breadth and original mechanism studies
R133	Source/correctness repair	Fourteen exposed qualification positions and four old/new S2 bridge pairs
R134	S2 with independent zero finalization	Final confirmation: 32 method positions; four new weight runs plus two reused points
R135	Same S2, optional AP row	Two incremental LPs; preregistered 10-arm first stage and conditional 22-arm second stage

The R134 confirmation uses opposite-order micro repetitions at Seeds 0/1 (90-second cap); K-15-S (600 seconds); UC-B20-B (1,200 seconds); and K-35-U, X-35-U and X-35-B (2,400 seconds). P and S2 appear at every paired role, with PS additionally present for K-15-S and X-35-B. Four S2 runs at $\lambda = .075, .30$ supplement the fixed .15 Seed0 baselines for K-15-S and UC-B20-B. The supplement gives every position, cap, Seed, repeat and measured cost.

An open run has no certification time. For two certified methods, a time difference is substantive when

$$t_{\text{slow}} - t_{\text{fast}} \geq \max\{10\text{ s}, 0.10 t_{\text{fast}}\}.$$

Micro observations retain their absolute repeated times. Open comparisons retain both U and L, the signed gap $U - L$, and source qualification; better primal and weaker dual endpoints are stated separately. Historical clocks are descriptive across environments, whereas each matched comparison shares its timing environment.

Coordinate sources and generated operational data are separate. CITI instances use the archived 443-station Citi Bike coordinate/capacity table; CABI instances use the archived Capital Bikeshare station table, and X instances use the archived Chicago Divvy station snapshot. Selection manifests identify source rows, hashes and projected coordinates. Depots are synthetic centroids; inventories, positive targets and weights follow recorded generation rules. D13 is entirely synthetic; the micro regression input uses CABI coordinates with generated operational data. These inputs test a static operational model and do not reconstruct observed trips or live station demand. Provider data services are attributed separately from the archived snapshot identities (Citi Bike 2026; Capital Bikeshare 2026; Divvy 2026). The supplement records the source-to-input chain and the specific source-capture facts still needing author confirmation.

5. Complete-method and operational results

5.1 Historical breadth and attribution

The R132 twelve-input/two-Seed validation has 24 S2 and 24 P observations and 12 PS observations. Its original qualified certificate counts are 17/24, 9/24 and 4/12. S2/P has 16 favorable time/certificate directions, one adverse direction, six both-open comparisons and one comparison ineligible under its source policy. The excluded S2 observation has its own legal zero fleet but no production certificate qualified by that historical policy. These counts remain historical; the new independent-proof rule does not rewrite them.

Earlier six-role final-version evidence and the support-policy study show useful complete-method performance, but PS reversals matter. PS and inclusive-root controls establish that an initial feasible solution does not explain every S2 gain. They do not isolate a statistically or practically strong independent speed benefit from the outer controller. Equivalent-expression controls have mixed cost effects. S2 is supported as an integrated method; component-level causality requires its corresponding matched controls.

The R133 source/correctness bridge and its complete qualification record remain separately identified in the supplement. They do not relabel earlier performance observations. The current AP comparison uses the accepted S2 baseline and changes only the aggregate row.

5.2 Final-version confirmation

The final confirmation produces 12/14 S2, 9/14 P and 2/4 PS certificates. Against cold P, S2 has nine both-certified comparisons and three additional certificates; two pairs remain open on both sides. A repeated advantage on the actual split/child-target role UC-B20-B saves about 315 seconds at each Seed. X-35-U includes a large complete-time improvement, and all micro repetitions have lower S2 absolute time.

Paid-start compact remains effective on X-35-B: PS certifies in about 7.2 seconds while S2 takes about 18 seconds at both Seeds. Both reversals meet the substantive-difference rule. In both R134 S2 runs the own startup objective is positive, 0.0502575215; the zero fleet is obtained later. The reversal therefore concerns the complete search path and does not indicate that S2 began with an already optimal zero startup fleet. Table 2 retains them alongside the S2 gains. One independently certified exact-zero observation retains its rejected native-source flag.

Table 2. All final-panel matched groups. C is a certificate with complete production seconds; O reports feasible U / qualified L and has no certificate time. Seed and repeat follow the problem name. Displayed values are rounded; Supplement S1-S2 retains full-precision values.

Problem / Seed / repeat	P	PS	S2	S2/P result
SM-B10-B / 0 / 1	C 1.641s	-	C 0.857s	S2 faster (micro)
SM-B10-B / 0 / 2	C 1.041s	-	C 0.774s	S2 faster (micro)
SM-B10-B / 1 / 1	C 1.043s	-	C 0.805s	S2 faster (micro)
SM-B10-B / 1 / 2	C 1.016s	-	C 0.773s	S2 faster (micro)
K-15-S / 0 / 1	O 0.59226/0.52741	O 0.59226/0.52022	C 112.935s	S2 only certified
K-15-S / 1 / 1	O 0.59226/0.5289	O 0.59226/0.53187	C 115.021s	S2 only certified
UC-B20-B / 0 / 1	C 319.060s	-	C 3.966s	S2 faster; substantive
UC-B20-B / 1 / 1	C 318.377s	-	C 3.821s	S2 faster; substantive
K-35-U / 0 / 1	O 0.66475/0.3355	-	O 0.69285/0.56494	both open
K-35-U / 1 / 1	O 0.75885/0.33055	-	O 0.66836/0.55427	both open
X-35-U / 0 / 1	O 0.016265/6.9389e-18	-	C 120.257s (ind.)	S2 only certified
X-35-U / 1 / 1	C 2064.949s (ind.)	-	C 452.841s (ind.)	S2 faster; substantive
X-35-B / 0 / 1	C 157.954s (ind.)	C 7.251s (ind.)	C 17.606s (ind.)	S2 faster; substantive
X-35-B / 1 / 1	C 50.151s (ind.)	C 7.196s (ind.)	C 18.277s (ind.)	S2 faster; substantive

The full per-arm table in the supplement includes every micro repetition, certificate class, endpoint and complete production/verification cost. Source-rejected native conclusions remain rejected even where an independent exact-zero certificate exists. Actual split, child-target and cache observations describe exposed controller behavior rather than proving an independent controller speed effect. No natural NEXT observation is required to complete this finite confirmation.

5.3 Size and physical interpretation

The unresolved K-35-U endpoints delimit the positive-objective size evidence. The complete endpoint tables retain both bounds at each Seed. At Seed0 cold P has the better primal endpoint while S2 has the better lower bound; at Seed1 S2 improves both endpoints. This differs from the certified zero objectives on X-35-U and X-35-B: station count alone does not describe proof difficulty. K-35-B's optimum class remains unconfirmed.

Physical operation tables retain all quantities, final stocks, returns, prefix peaks and durations. A legal historical fleet collects 58 units with capacity 20 by delivering at intermediate stations; cumulative pickup remains compatible with prefix capacity. Loaded returns remove stock from station inventories and incur depot unloading time.

5.4 Within-instance objective weights

For a fixed physical feasible set and $0 < \rho_a < \rho_b$, let x_a, x_b be globally optimal. Write $\Delta P = P_{\text{avg}}(x_a) - P_{\text{avg}}(x_b)$ and $\Delta G = G(x_b) - G(x_a)$. The two optimality inequalities give

$$\Delta P \geq 0, \qquad \rho_a \Delta P \leq \Delta G \leq \rho_b \Delta P.$$

Indeed, optimality at ρ_a gives the lower inequality, optimality at ρ_b gives the upper one, and their compatibility yields $(\rho_b - \rho_a)\Delta P \geq 0$. Increasing the penalty weight therefore weakly reduces average relative deviation and weakly increases Gini among global optima; a platform and multiple optima are allowed.

If only feasible solutions with qualified objective errors $\varepsilon_a, \varepsilon_b \geq 0$ are known, the justified statements are

$$\rho_a \Delta P - \varepsilon_a \le \Delta G \le \rho_b \Delta P + \varepsilon_b, \qquad \Delta P \ge -\frac{\varepsilon_a + \varepsilon_b}{\rho_b - \rho_a}.$$

Here errors must be supported by valid original-objective upper/lower bounds and their numerical policy. A primal feasibility tolerance is not an objective-error bound.

All six points are certified. On K-15-S, increasing the weight reduces average relative deviation and increases Gini; pickups rise from 26 to 28 units, and the longest route uses over 99% of its duration budget. On UC-B20-B, all three weights give the same observed Gini/deviation values and aggregate operations: 14 pickups and 6 units returned. This platform is consistent with the optimality relations above. Effective weights and full-precision endpoints accompany the vehicle records; accepted objective gaps, rather than feasibility tolerance, control the numerical interpretation.

Figure 1 locates all six measured points. Connecting segments guide the eye and do not estimate an unobserved optimal frontier.

Table 3. Fixed physical instances at all three weights. Gap is signed original-objective U-L.

Problem	lambda / rho	G	P average	F	certificate	gap
K-15-S	0.075 / 0.710503	0.20237	0.276702	0.398967	yes	5.55112e-16
K-15-S	0.15 / 1.42101	0.222147	0.260462	0.592265	yes	3.01757e-09
K-15-S	0.3 / 2.84201	0.225433	0.258588	0.960344	yes	2.22045e-16
UC-B20-B	0.075 / 0.88725	0.170833	0.229079	0.374083	yes	1.11022e-16
UC-B20-B	0.15 / 1.7745	0.170833	0.229079	0.577333	yes	2.22045e-16
UC-B20-B	0.3 / 3.549	0.170833	0.229079	0.983833	yes	4.44089e-16

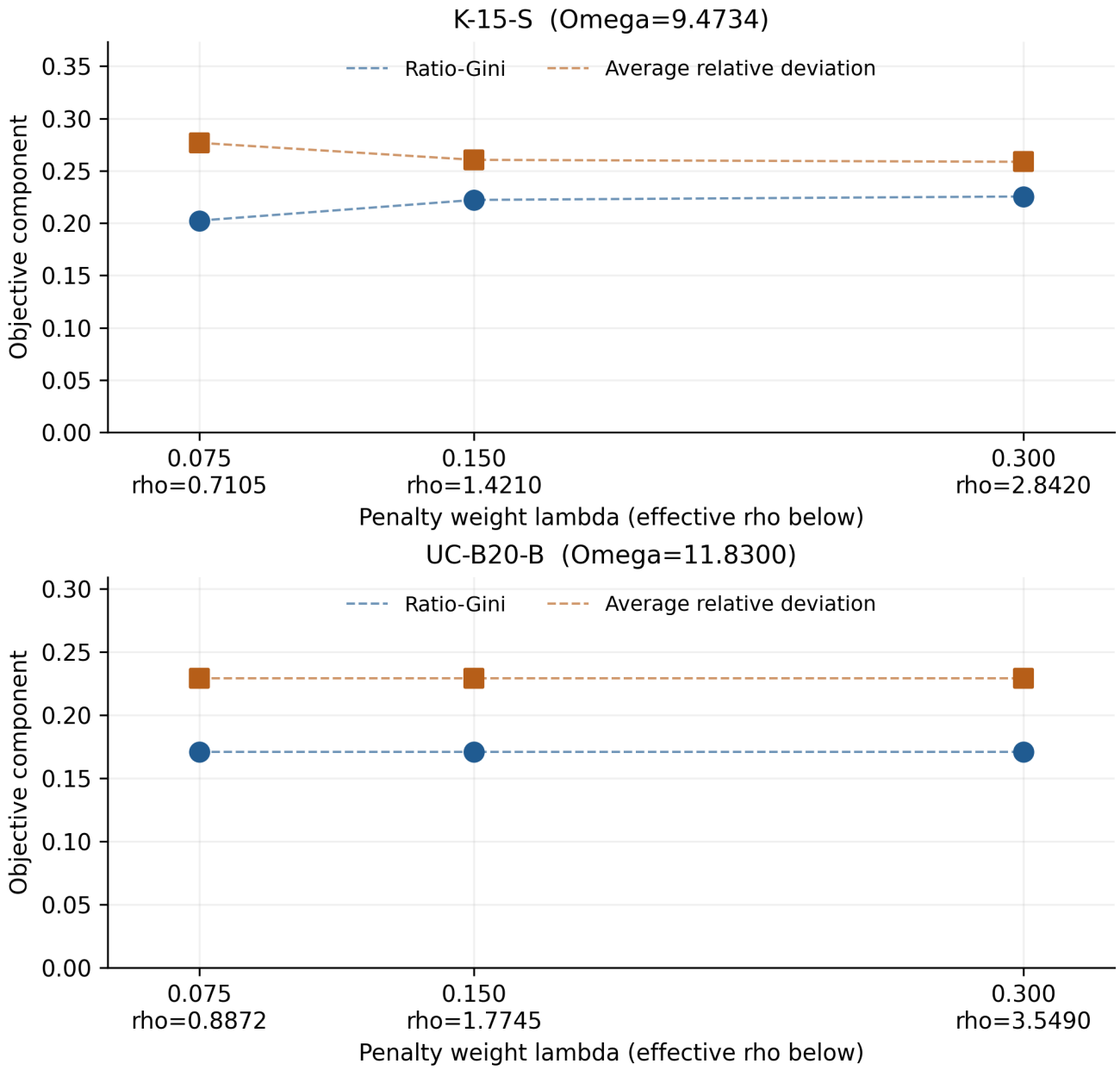


Figure 1: Within-instance changes in ratio-Gini and average target deviation. Labels identify lambda; filled symbols are certified observations and open symbols are feasible uncertified observations.

6. State information, relaxation strength and complete search

An objective epigraph can discard information retained in a finite inventory mixture. Let $\rho_{iy} = y/D_i$ in the ideal model and define

$$\phi_i(s) = \sum_y |\rho_{iy} - 1| s_{iy}, \qquad a_i^+ = \sum_y (\rho_{iy} - 1)_+ s_{iy}, \quad a_i^- = \sum_y (1 - \rho_{iy})_+ s_{iy}.$$

Since $r_i - 1 = a_i^+ - a_i^-$, the state penalty exceeds the absolute deviation of its mean by

$$\phi_i(s) - |r_i - 1| = 2 \min\{a_i^+, a_i^-\} \geq 0.$$

This is the familiar convexity loss under mixing (Boyd and Vandenberghe 2004). States on opposite sides of the target can cancel in the original epigraph; one-sided support gives no such loss. The diagnostic does not itself equal the improvement after reoptimization.

The historical VD-J formulation already tested stationwise joint penalty equalities. It reached root, pilot and development studies, but its stronger root bounds did not prevent a certificate loss and severe MIP regression on D13. The later S2 baseline has a shared inventory/service representation. We therefore test only the aggregate lower bound

$$\sum_i \omega_i e_i \geq \sum_i \omega_i \phi_i(s), \tag{AP}$$

which adds one row and no variables. The total nonzero count remains comparable to the stationwise system because the row is dense; fewer rows alone do not imply lower solve cost. Write $P_e = \sum_i \omega_i e_i$, and let x collect all variables except e. In the ideal formulation, if every applicable local lower bound is at most $\phi_i(s)$, $\phi_i(s) \leq M_i$ for the actual finite local upper bound M_i, and every other use of e is through its weighted total, AP has the same (x, P_e) projection as all stationwise lower bounds $e_i \geq \phi_i(s)$. The supplement gives the constructive proof, handles zero weights and states why this does not imply equivalence to VD-J equalities or independent Se_i products.

The two fixed S2 root domains confirm the intended numerical effect. K-20-S increases from 0.1951491524 to 0.3187585426; K-35-U increases from 0.2001776619 to 0.2738576813. AP matches the previously observed stationwise optima within the existing numerical policy. All layers use the same input, support, interval and cutoff. The original and stationwise optima are reused; exactly two new continuous LPs add only AP. Production coefficients use the loaded binary64 ratio coefficients and downward-rounded nonnegative state coefficients. The ideal projection result and floating-point observations are distinct statements.

All 32 preregistered positions completed, with 10/16 certificates for each method. K-20-S remains open with the same U at both Seeds; AP reduces the gaps by about 7% and 30%. K-35-U also remains open: AP improves U at both Seeds, but its qualified L is weaker at both. Its gaps are smaller by about 25% and 2.8%, so the result is a primal/dual trade-off rather than endpoint dominance or faster proof. On the original D13 input both methods remain open at the same U. AP slightly narrows the Seed0 gap but widens the Seed1 gap by about 14%. This is mixed protection evidence, despite the absence of a new certificate loss.

Among the ten both-certified pairs, AP has no substantive time advantage and two substantive adverse differences: K-15-S Seed0 takes 149.31 seconds versus 112.11 for S2, and X-35-B Seed0 takes 79.97 versus 17.59. Their Seed1 differences are small. UC-B20-B remains certified in about four seconds on both sides, and micro repetitions have small absolute differences. Most of the two slowdowns lies in native MIP time, accompanied by

higher Work and node counts; LP setup is a small part of the added cost. Complete production, postverification and total clocks, every endpoint and the exposed root/child/MIP costs are retained in Supplement S4.

A separate inventory-threshold count inequality has a strict exact increment of $3/16$ on two named tiny joint-service references while adding no reliable increment on these real roots after retaining the old state-penalty control. Its full derivation, controls and negative real-root evidence remain in the supplement. These comparisons distinguish useful state information, stronger relaxations and faster complete algorithms.

7. Conclusions and implications

Shared inventory/service states provide a coherent exact formulation for ratio-Gini bicycle redistribution. Unique one-way service yields implicit service integrality; ordered supply and support/handling information connect attainable inventory to vehicle resources. Complete interval coverage and physical verification turn qualified native bounds into whole-problem evidence.

The supported S2 implementation improves certification coverage and complete runtime against cold compact on the stated historical and final panels. Independently paid-start compact reversals and open positive-objective 35-station cases remain meaningful limits. The weight experiment shows why normalized penalty weight and actual vehicle operations are needed to interpret equitable service restoration.

AP retains the intended numerical root strengthening and improves some complete-search endpoints, but it does not improve certification coverage or establish a reliable complete-time advantage. The substantial Seed0 slow-downs and mixed D13 protection outcome do not justify replacing the accepted S2 method. We retain AP as a default-off research option and keep S2 as the supported main algorithm. The projection result explains retained state information under its stated premises; it does not guarantee a faster native search.

The principal limitations are targeted rather than random validation inputs, synthetic operating data on geographic coordinate sources, a single qualified native backend, and unresolved positive-objective scale. Future evidence should address these boundaries while retaining the full cost and proof-source distinctions used here.

Appendix A. Complete model, scoped strengthening and proof obligations

The complete physical formulation and strengthening proofs below are retained from the maintained manuscript. S2 uses uniform pair support and no triple support. The S2 termination rule includes the independent exact-zero proof in Section 3.4; native-source qualification and all normal positive-objective rules remain separate. Historical source pointers identify their own snapshots and are not claims that an old executable produced the final panel.

This appendix is the single maintained model specification for the manuscript. It transcribes the frozen writer, including constraints sometimes redundant for integer solutions. It does not replace the implemented model by a different vehicle-routing formulation. Equations are mathematical real-arithmetic expressions; A.7 states the separate numerical implementation boundary. Historical source-line pointers retained in this derivation refer to snapshot 355eefd2be3fe5651195681faf7b3a63ee4032f8. That historical E3 source base and its binding belong to the earlier attribution controls (TailoredExact project 2026). The R134 reference identities and the R135 default-off AP recompilation are distinguished in B.1; source_publication.json provides the current exact source-byte capsule. Current state/service, own-H, pair-only and inclusive-domain conditions are checked against these final identities, rather than inferred from historical line numbers.

A.1 Sets, data and variables

Let $N = \{1, \dots, n\}$ be service stations, $N_0 = N \cup \{0\}$, $K = \{1, \dots, m\}$, and $A = \{(i, j) \in N_0^2 : i \neq j\}$. Depot 0 is absent from every inventory, ratio, weight and Gini sum. Vehicle indexing is shifted by one relative to source indices $0, \dots, M-1$. All vehicle-indexed formulas hold for each $k \in K$; all station formulas hold for each $i \in N$, unless otherwise stated.

Station capacity C_i and initial inventory b_i are integers with $0 \leq b_i \leq C_i$; target inventory D_i is a positive integer no larger than C_i. The parser's depot capacity/stock fields do not provide departure supply. Vehicle capacity $Q_k > 0$ is integer. A vehicle starts empty and returns to the same depot, possibly loaded. Travel time $\tau_{ij} = \|v_i - v_j\|_2/1.5$ uses the serialized planar coordinates. The route horizon is T; unit pickup/drop times are t_p, t_d, with $h = t_p + t_d$. Penalty multiplier $\lambda \geq 0$ and station weights $\omega_i \geq 0$ are data; the baseline panel uses $\lambda = .15$, with within-instance sensitivity at $.075, .30$, positive weights and $t_p = t_d = 60$. No inventory floor is imposed by the compatibility field min_ratio.

The binary variables are arc decisions x_{kij}, station service assignments z_{ki}, and pickup-direction indicators a_{ki}. Source calls the last variable mName(k,i); we rename it a to avoid confusing it with the fleet size. Pickup p_{ki}, drop d_{ki}, final inventory Y_i, and post-service load ℓ_{ki} are nonnegative. The rank u_{ki} is continuous. Define $\bar{p}_{ki} = \min(b_i, Q_k)$ and $\bar{d}_{ki} = \min(C_i - b_i, Q_k)$.

Variable	Domain and bound in the active formulation
x_{kij}, z_{ki}, a_{ki}	Binary, for all stated indices; no self arcs
p_{ki}, d_{ki}	Integer for cold P and ENS-C; continuous for M-B; upper bounds $\bar{p}_{ki}, \bar{d}_{ki}$
ℓ_{ki}	Integer in $[0, Q_k]$, retained by all three methods
u_{ki}	Continuous in $[0, n]$
Y_i	Integer in $[\underline{y}_i, \bar{y}_i]$; plain model has $[0, C_i]$
r_i	Continuous in $[0, C_i/D_i]$
e_i	Continuous in $[0, \max(1, C_i/D_i - 1)]$
$H_{ij}, i < j$	Continuous in $[0, C_i/D_i + C_j/D_j]$
g, Z_i	Continuous; $g \in [0, 1]$ in the unsliced writer, $g \in [\alpha, \beta] \subseteq [0, 1]$ for an interval; $Z_i \in [0, C_i]$
f_{kij}	Continuous in $[0, n]$, F0 connectivity flow only

Variable	Domain and bound in the active formulation
s_{iy}, q_{iy}	Binary inventory selectors and continuous selected-G products, for $y \in \mathcal{Y}_i = \{\underline{y}_i, \dots, \bar{y}_i\}$, VD-P only
$b_{it}^{\mathrm{bit}}, w_{it}$	Binary inventory bits and continuous products, plain P only
W_{SP}	Continuous in $[\max(0, \underline{SP}), \max\{\max(0, \underline{SP}), \bar{S}\bar{P}\}]$, when the aggregate SP estimator is active

The model objective is $f = g + \lambda P$, where $P = \sum_i \omega_i e_i$; $f, S = \sum_i r_i$, and $H_\Sigma = \sum_{i<j} H_{ij}$ are expressions, not new columns. Constants $\underline{y}, \bar{y}, \underline{S}, \bar{S}, \underline{P}, \bar{P}$ are defined in A.5. Bound-only extrema columns r_min,r_max may still be allocated by the writer; with the shared factory their centering inequalities use the projected pair form below and those columns have only their original bounds $[0, \max_i C_i/D_i]$. No additional vehicle-used binary exists: $v_k = \sum_j x_{k0j}$ is an expression.

A.2 Routing, service and physical resources

The complete routing core is

$$\sum_{j \in N} x_{k0j} = \sum_{i \in N} x_{ki0}, \qquad \sum_{j \in N} x_{k0j} \le 1, \tag{A1}$$

$$\sum_{j \in N_0 \setminus \{i\}} x_{kji} = \sum_{j \in N_0 \setminus \{i\}} x_{kij} = z_{ki}, \qquad \sum_k z_{ki} \le 1, \tag{A2}$$

$$z_{ki} \le u_{ki} \le n z_{ki}, \qquad u_{ki} - u_{kj} + n x_{kij} \le n - 1 \quad (i, j \in N,\ i \ne j). \tag{A3}$$

These are the writer's original depot degree, station degree and MTZ rows. Any selected station cycle contradicts the sum of (A3) around the cycle. Thus an integer nonempty selection has one depot-connected elementary tour; an unused vehicle has all arcs and assignments zero. A depot-to-depot empty route is represented without a self arc. The MTZ rows remain when F0 also adds the following connectivity flow:

$$0 \le f_{kij} \le n x_{kij} \quad ((i,j) \in A), \qquad \sum_{j \ne i} f_{kji} - \sum_{j \ne i} f_{kij} = z_{ki}, \tag{A4}$$

$$\sum_{j \in N} (f_{k0j} - f_{kj0}) = \sum_{i \in N} z_{ki}.$$

F0 here means the round20-current flow variant, including return-arc columns, with no $f \ge x$ normalization or start-coupling additions. A route visiting q stations admits downstream-count flows $q, q-1, \dots, 1, 0$. These flows establish validity for every original route, independently of MTZ. Cold plain P does not receive this strengthening. The row package also called F0 has additional families in A.5; these two uses of the historical label must not be confused.

All protected service rows are retained:

$$p_{ki} \le \bar{p}_{ki} z_{ki}, \quad d_{ki} \le \bar{d}_{ki} z_{ki}, \quad a_{ki} \le z_{ki}, \quad p_{ki} \le \bar{p}_{ki} a_{ki}, \tag{A5}$$

$$d_{ki} \le \bar{d}_{ki}(z_{ki} - a_{ki}), \quad p_{ki} + d_{ki} \ge z_{ki}, \quad 0 \le \ell_{ki} \le Q_k z_{ki}.$$

The first pair is not silently dropped even where implied by direction rows. Zero directional bounds cause no division. Unique service, a binary direction and the last service inequality forbid split service, simultaneous pickup/drop and zero-operation visits.

Loads are the writer's original two-sided Big-M recurrences:

$$-Q_k(1 - x_{k0i}) \le \ell_{ki} - p_{ki} + d_{ki} \le Q_k(1 - x_{k0i}), \tag{A6}$$

$$-Q_k(1 - x_{kij}) \le \ell_{kj} - \ell_{ki} - p_{kj} + d_{kj} \le Q_k(1 - x_{kij}) \quad (i, j \in N,\ i \ne j), \tag{A7}$$

$$R_k := \sum_i (p_{ki} - d_{ki}) \ge 0, \qquad \sum_{(i,j)\in A} \tau_{ij} x_{kij} + h \sum_i p_{ki} \le T. \tag{A8}$$

The first selected arc fixes departure load to zero. Selected internal arcs propagate each post-service load. A return has no service operation: its load equals the last ℓ, hence is at most Q_k. Summing the recurrence gives R_k as that load, so its upper bound follows without a separate R_k column or explicit upper row. Inactive recurrence rows are valid because their difference is between a feasible before-service load and another feasible post-service load, each in $[0, Q_k]$. Empty vehicles have $p = d = \ell = R = 0$.

In (A8), let $P_k = \sum_i p_{ki}$ and $D_k^{\text{drop}} = \sum_i d_{ki}$. Since $R_k = P_k - D_k^{\text{drop}}$, original handling time is

$$t_p P_k + t_d(D_k^{\text{drop}} + R_k) = (t_p + t_d)P_k. \tag{A9}$$

Return unloading is paid. Capacity Q_k limits every load, not cumulative pickup P_k, which may exceed Q_k after intermediate drops. There is one physical time limit per vehicle and a separate computational deadline.

Inventories and objective auxiliaries satisfy

$$Y_i + \sum_k (p_{ki} - d_{ki}) = b_i, \quad r_i = Y_i/D_i, \quad e_i \ge r_i - 1, \quad e_i \ge 1 - r_i, \tag{A10}$$

$$H_{ij} \ge r_i - r_j, \quad H_{ij} \ge r_j - r_i \quad (i < j). \tag{A11}$$

Consequently $\sum_i Y_i + \sum_k R_k = \sum_i b_i$. Loaded returns permit a decrease of station inventory; no external depot inventory can increase it.

A.3 Original and disaggregated Gini objective

The physical objective is

$$F(Y) = G_{\text{true}}(Y) + \lambda \sum_i \omega_i |Y_i/D_i - 1|, \quad G_{\text{true}} = \begin{cases} \dfrac{\sum_{i<j} |r_i - r_j|}{n \sum_i r_i}, & S > 0, \\ 0, & S = 0. \end{cases} \tag{A12}$$

The native epigraph is

$$n \sum_i Z_i/D_i \ge H_\Sigma, \qquad \min f = g + \lambda \sum_i \omega_i e_i. \tag{A13}$$

Cold P uses inventory bits. With $B_i = \max(1, \lceil \log_2(C_i + 1) \rceil)$, for $t = 0, \ldots, B_i - 1$,

$$Y_i = \sum_t 2^t b_{it}^{\text{bit}}, \quad Z_i = \sum_t 2^t w_{it}, \quad 0 \le w_{it}, \quad w_{it} \le g, \quad w_{it} \le b_{it}^{\text{bit}}, \quad w_{it} \ge g + b_{it}^{\text{bit}} - 1. \tag{A14}$$

The capacity bound removes unused bit encodings. These products are exact at binary bits. ENS-C and M-B instead use the finite value-disaggregated perspective block VD-P:

$$\sum_{y\in\mathcal{Y}_i} s_{iy} = 1, \quad Y_i = \sum_y y s_{iy}, \quad \alpha s_{iy} \le q_{iy} \le \beta s_{iy}, \quad \sum_y q_{iy} = g, \quad Z_i = \sum_y y q_{iy}. \tag{A15}$$

The active mode is VD-P, not VD-J: equalities reconstructing r_i or e_i from selectors are not substituted for (A10). The shared factory's bit-product McCormick rows are skipped for this block because those bit columns are absent. Products (A15) already use interval bounds. M-B additionally writes the original A/B rows

$$\sum_k (p_{ki} + d_{ki}) = \sum_{y\in\mathcal{Y}_i} |b_i - y| s_{iy}, \qquad \sum_k z_{ki} + s_{i,b_i} = 1. \tag{A16}$$

A missing initial state means $s_{i,b_i} = 0$, not a new selector. The independent ENS-Q option is false. The mode-linked direction equality of another variant is absent; M-B retains its original binary a_{ki}. ENS-C uses integer p/d and omits (A16). These are the material representation differences; the complete H and controller are shared.

A.4 Integrality, embedding and fairness propositions

Lemma A1 (implicit service integrality). With integer b_i, Y_i, (A2), (A5), and binary assignments/directions, M-B's continuous quantities are integer.

Proof. If $z = 0$, its quantities and direction are zero. At the only assigned vehicle, direction 1 forces $d = 0$, so $p = b_i - Y_i$ is integer. Direction 0 forces $p = 0$, so $d = Y_i - b_i$ is integer. No other vehicle contributes. Retained integer loads are not needed for this argument. Removing unique service, one-way direction or integer inventory invalidates the inference. The statement concerns integer feasible solutions, not equality of LP convex hulls. □

Validity of A/B. For an integer original service, total movement at a station equals $|b_i - Y_i|$, which its selected state reproduces. An assigned service is nonzero and one-way, so $Y_i \neq b_i$ exactly when the station is served. This proves both (A16); neither is an extra premise of Lemma A1.

Lemma A2 (physical objective embedding). Assume inventory domains and rows preserve all physical points assigned to a true-G interval $[\alpha, \beta]$ and cutoff $F \le c$. Every such fleet has an extension satisfying the applicable original or VD-P groups, with $g = G_{\text{true}}$ and $f = F$, up to permutation of identical vehicle labels if symmetry rows are present.

Proof. Use the physical arcs, service, load prefixes, tour ranks and downstream flows. Select its integer inventory state, put $q_{iy} = g$ at the selected state and zero at others; (A15) yields $Z_i = gY_i$. Set $e_i = |r_i - 1|$ and $H_{ij} = |r_i - r_j|$. Then (A13) holds, with equality for $S > 0$. At $S = 0$ all inventories, ratios and differences are zero and $g = 0$ realizes the stipulated convention. Bounds and valid strengthening are satisfied by assumption. Conversely an integer product completion has $Z_i = gY_i$, hence $g \ge G_{\text{true}}$ for $S > 0$ and $f \ge F$; at zero sum, $g \ge 0$ suffices. □

Arbitrary native g may exceed physical Gini. A fleet with $G_{\text{true}} < \alpha$ can have an inflated-g auxiliary completion. The proof assigns physical points by their true Gini and requires their objective-preserving extension; it does not classify every completion by its auxiliary g. An inventory vector without compatible routes is not an upper-bound witness.

For $n > 1$, triangle inequalities give $A := \sum_{i<j} |r_i - r_j| \le (n-1)S$, so $G_{\text{true}} \le (n-1)/n$. For $S > 0$, $G_{\text{true}} = 0$ means every ratio is equal; at $S = 0$ all are zero by nonnegativity. A common ratio below one is a common proportional shortage, not target attainment. If $\lambda > 0$ and all $\omega_i > 0$, $F = 0$ if and only if every $Y_i = D_i$. These are dispersion and target-deviation statements, with no implied demographic/service/revenue interpretation.

A.5 Enabled strengthening: construction, proofs and scope

The finite list below describes the actual M-B path, not every option in the repository. All rows are assembled **statically when a canonical interval model is written**, before its LP or MIP request. A new cutoff epoch rewrites the relevant model; it does not separate these rows at fractional native nodes. The writer supplies physical rows, and the shared interval factory supplies the interval rows. Families registered by the factory are not all rows emitted by it. In particular, the factory's bit-product rows are filtered out for VD-P. The original objective and physical constraints remain present.

A.5.1 Construction of the inventory rectangle (D1)

Let the current true-G interval be $I = [\alpha, \beta]$, the independently verified own incumbent be U_e, and the cutoff be $c = U_e$. **The active controller sets incumbent_epsilon to zero**, so this is a non-strict objective cutoff; a generic writer can subtract an epsilon, but that is not this frozen request. Start with $l_i = 0, u_i = C_i$. With $\lambda > 10^{-12}$, finite positive U_e, and $B = (c - \alpha)/\lambda \geq -10^{-10}$, for each $\omega_i > 10^{-12}$ intersect with

$$l_i \leftarrow \max\{l_i, \lceil D_i(1 - \max(0, B/\omega_i)) - 10^{-9} \rceil\}, \quad u_i \leftarrow \min\{u_i, \lfloor D_i(1 + \max(0, B/\omega_i)) + 10^{-9} \rfloor\}. \tag{A19}$$

This is followed by one intersection with physical movement reachability. Write $t_i = \tau_{0i} + \tau_{i0}$ and

$$\begin{aligned} M_{ki} &= \begin{cases} \min\{Q_k, \lfloor (T - t_i)/h + 10^{-9} \rfloor\}, & T + 10^{-9} \geq t_i,\ h > 10^{-12}, \\ Q_k, & T + 10^{-9} \geq t_i,\ h \leq 10^{-12}, \\ 0, & \text{otherwise.} \end{cases} \\ l_i &\leftarrow \max\{l_i, b_i - \max(0, \max_k \min(b_i, Q_k, M_{ki}))\}, \\ u_i &\leftarrow \min\{u_i, b_i + \max(0, \max_k \min(C_i - b_i, Q_k, M_{ki}))\}. \end{aligned} \tag{A20}$$

The outer max with zero transcribes the actual zero-initialized reachability accumulators, including a slightly negative quotient admitted by a floating travel guard. The mathematical version omits the floating guards, uses $h = 0$ for the second case, and clips movement below at zero. Positive published $h = 120$ avoids the small-positive-h threshold ambiguity. If an interval becomes empty, the code flags domain infeasibility, resets that station's storage bounds to $[0, C_i]$, and emits an impossible row. The reset prevents malformed column ranges; it does not reinstate the excluded domain as feasible.

Proposition A5 (budget and reachability). Every physical point in $I, F \leq c$ satisfies the ideal versions of (A19)-(A20). Indeed $F = G + \lambda P$, $G \geq \alpha$, and nonnegative weighted deviations imply $\omega_i |Y_i/D_i - 1| \leq (c - \alpha)/\lambda$. Integer Y_i gives the rounded bounds. A visited station requires a depot tour with travel at least t_i. Its pickup is at most Q_k, b_i, and its drop is at most $Q_k, C_i - b_i$; because $D_k^{\text{drop}} \leq P_k$, either operation is at most cumulative pickup, which has handling cost $hP_k \leq T - t_i$. Maximize over its unique possible vehicle. These are necessary, not sufficient, attainability conditions: a delivery still needs a source. Zero weight or zero lambda supplies no individual penalty bound and is skipped. □

Actual order and pass count. The writer first computes budget bounds, intersects reachability, and derives interval extrema; the shared factory recomputes this same sequence from $[0, C_i]$ and replaces the local arrays. It does not feed the first result into a second tightening pass. There is no propagation loop in this enabled path and no iterative fixed point. The preset advertises `iterative` and requested rounds 2, but the legacy completion field is capped at 1. Thus the operative definition is one budget intersection plus one reachability intersection,

independently recomputed by the factory. R124's "original two passes" description was incorrect; production is unchanged.

Set $\underline{y}_i = l_i, \bar{y}_i = u_i, \mathcal{Y}_i = \{l_i, \dots, u_i\}$, and

$$\begin{aligned} &\underline{r}_i = l_i/D_i, \quad \bar{r}_i = u_i/D_i, \\ &\underline{S} = \sum_i \underline{r}_i, \quad \bar{S} = \sum_i \bar{r}_i, \\ &\underline{e}_i = \operatorname{dist}(1, [\underline{r}_i, \bar{r}_i]), \quad \bar{e}_i = \max\{|\underline{r}_i - 1|, |\bar{r}_i - 1|\}, \\ &\underline{P} = \sum_i \omega_i \underline{e}_i, \quad \bar{P} = \sum_i \omega_i \bar{e}_i. \end{aligned} \tag{A21}$$

The last extrema bound the **tight physical completion** of e. They do not assert that every nonminimal native epigraph e_i is at most $\bar{e}_i$. There is no stock-weighted knapsack or optimization in the denominator calculation: these are sums over the rectangle, with no extra conservation-driven tightening.

A.5.2 Physical conservation and activation (P1-P4)

The following rows hold on the full physical domain, independently of (I,c):

$$\sum_i b_i - \sum_k Q_k \le \sum_i Y_i \le \sum_i b_i; \quad -b_i \sum_k z_{ki} \le Y_i - b_i \le (C_i - b_i) \sum_k z_{ki} \quad (i \in N); \tag{A22}$$

$$h \sum_{k,i} p_{ki} \le mT; \qquad z_{ki} \le 0 \quad \text{if } t_i + h > T + 10^{-9}. \tag{A23}$$

Proof. Each return load lies in $[0, Q_k]$; insert this into conservation (A10). If no vehicle serves i, inventory balance fixes $Y_i = b_i$; otherwise unique assignment makes the visit sum one and the two activation bounds reduce to $0 \le Y_i \le C_i$. Sum (A8) and drop nonnegative travel for global handling. Any nonzero integer service has at least one pickup somewhere on the vehicle: for a drop use empty departure and load nonnegativity. Consequently a tour through i takes at least $t_i + h$. The physical meaning of the historical `transfer_subset_capacity` flag here is this singleton visit exclusion, not a separate general subset-flow formulation. All vehicles and stations are enumerated in increasing index order. These inequalities are implied for integer physical points, but can strengthen the relaxation. The global-handling impossible-row diagnostic based on accumulated required pickup is inactive in the shared-factory path because that legacy accumulator is not populated there. □

A.5.3 Ordered supply compatibility (P5)

For every vehicle k and receiving station j, define

$$\mathcal{C}_{kj} = \{i \in N \setminus \{j\} : \tau_{0i} + \tau_{ij} + \tau_{j0} + h \le T + 10^{-9}\}. \qquad d_{kj} \le \sum_{i \in \mathcal{C}_{kj}} p_{ki}. \tag{A24}$$

All stations passing this test enter the set, even those with zero stock; their pickup column is then fixed at zero. Common T, h, τ make the sets identical across vehicles, but their pickup bounds still use Q_k. No demand assumption or target sign is used.

Proposition A6. (A24) is necessary for a physical fleet under metric travel and empty departure. Label the integer units collected by a vehicle. Scan its route in order, assigning each delivered unit to one previously collected, not yet delivered unit. Every prefix load is nonnegative, so this matching always exists. If a unit received at j came from i, the route includes the ordered subsequence (0,i,j,0). Repeated triangle inequalities give travel at least $\tau_{0i} + \tau_{ij} + \tau_{j0}$, and its at least one collected unit entails $hP_k \geq h$. Hence $i \in \mathcal{C}_{kj}$. Counting matched units at j proves the row; a pickup unit can be delivered only once. Intermediate stations can add travel or receive other units but cannot create supply. Unmatched collected units may return to depot. An unvisited source has $p_{ki} = 0$; an empty compatibility set forces $d_{kj} = 0$. Preloading, cross-vehicle handoffs, or nonmetric shortcut assumptions would break this proof and are excluded. □

The row is a coarse aggregate necessity, not a claim that all compatible pickups can simultaneously reach j. It neither imposes $P_k \leq Q_k$ nor substitutes for prefix loads.

A.5.4 Two- and three-station support (P6)

For $W = \{a, b\}$,

$$\tau(W) = \min\{\tau_{0a} + \tau_{ab} + \tau_{b0}, \tau_{0b} + \tau_{ba} + \tau_{a0}\}.$$

For $W = \{a, b, c\}$, take the minimum of the six expressions

$$\tau_{0\pi_1} + \tau_{\pi_1\pi_2} + \tau_{\pi_2\pi_3} + \tau_{\pi_3 0}, \quad \pi \text{ a permutation of } (a, b, c). \tag{A25}$$

These are explicit finite formulas, not new shortest-route optimization calls. Metric shortcutting of a complete tour yields one such order, so its travel is at least $\tau(W)$.

Lemma A7 (integer handling). If all stations in W are served by k, their nonzero integer operations give

$$|W| \leq \sum_{i \in W} (p_{ki} + d_{ki}) \leq P_k + D_k^{\text{drop}} \leq 2P_k, \quad P_k \geq \lceil |W|/2 \rceil. \tag{A26}$$

Here $P_k = \sum_{i \in N} p_{ki}$, not $\sum_{i \in W} p_{ki}$: supplies may lie outside W. Integer P_k gives the ceiling; the LP relaxation alone cannot justify it. Capacity does not bound cumulative pickup. Therefore the writer emits exactly one of the following rows for each selected support:

$$\begin{cases} \sum_{i \in W} z_{ki} \leq |W| - 1, & \tau(W) + h\lceil |W|/2 \rceil > T + 10^{-9}, \\ h \sum_{i \in W} p_{ki} + M_{kW} \sum_{i \in W} z_{ki} \leq T - \tau(W) + M_{kW}|W|, & \text{otherwise}, \end{cases}$$
$$M_{kW} = T + \tau(W) + h \max\{1, \sum_{i \in W} \min(b_i, Q_k)\}. \tag{A27}$$

Proposition A8 (both branches). The cover branch follows from (A8), (A25), (A26). In the conditional branch, if every station is served, cancel $M_{kW}|W|$. Since full travel is at least $\tau(W)$ and $\sum_W p \leq P_k$, the remaining inequality is necessary. If at least one is unserved, put $q = |W| - \sum_W z \geq 1$, $B_W = \sum_W \min(b_i, Q_k)$. The right side after moving the visit terms is $T - \tau(W) + qM_{kW} \geq 2T + h \max(1, B_W) \geq hB_W \geq h \sum_W p$. This proves deactivation for all nonnegative T, h, including $B_W = 0$, empty vehicle and $h = 0$. It does not rely on an invalid cumulative-capacity bound. □

Selection is finite and ordered. Current S2 visits vehicles k in increasing order and enumerates only increasing pairs $a < b$. Triple emission is disabled, so no triple row consumes the shared counter budget. One counter starts at zero outside the vehicle loop. The helper returns without adding a row once it reaches 50000; it increments

once before choosing cover versus conditional. At n=50, the current pair-only enumeration has 1225 candidate supports per full vehicle before this shared limit.

The historical full-M-B writer with triples enabled enumerates each vehicle's pairs first and then its increasing triples $a < b < c$. Only in that historical configuration does a full n=50 vehicle contribute 1225+19600=20825 supports, with the third vehicle receiving the prefix needed to reach 50000. Outer loops can continue evaluating cycle formulas after the helper stops adding rows. The limit counts emitted supports, rather than violated covers, and is shared across vehicles. Row validity is independent of selection policy; the dormant exhaustive-subset and rank3 variants have different coefficients/order and are excluded.

A.5.5 Dispersion, spread and centering (G1-G4)

For every pair $i < j$, let $d_{ij} = |r_i - r_j|$. For each $u \notin \{i, j\}$, triangle inequality gives $d_{iu} + d_{ju} \geq d_{ij}$. These pairs are distinct in the unordered numerator; the omitted pairs among the other stations are nonnegative. Hence

$$A = d_{ij} + \sum_{u \neq i,j} (d_{iu} + d_{ju}) + \sum_{u<v;\ u,v \neq i,j} d_{uv} \geq [1 + (n-2)]d_{ij} = (n-1)d_{ij}. \tag{A28}$$

For $n = 2$ the other sums are empty and equality holds. For $n = 1$, spread and centering are skipped. For true $G \in [\alpha, \beta]$ and $S > 0$, $n\alpha S \leq A \leq n\beta S$. At $S = 0$, all ratios and A are zero; only the interval containing true $G = 0$ needs that physical embedding. With **tight** $H_{ij} = d_{ij}$, the emitted rows are

$$n\alpha S \leq H_\Sigma \leq n\beta S; \qquad (n-1)H_{ij} \leq n\beta S \quad (i < j). \tag{A29}$$

This proves existence of an objective-preserving completion. It does **not** prove that an arbitrary inflated native H_{ij}, even at integer physical variables, satisfies the new row. For example $r = (1, 1), g = .5, H_{12} = 1$ satisfies the unsliced original epigraph $2gS \geq H$; the true-G-zero cap excludes this completion while preserving $H = 0, g = 0$. Accordingly these static rows are not advertised as callback user cuts preserving all original native integer completions.

Using the retained $H_{ij} \geq \pm(r_i - r_j)$ in the pairwise row gives the two variable-S rows, and $S \leq \bar{S}$ then gives the fixed rows:

$$(n-1)(r_i - r_j) \leq n\beta S, \quad (n-1)(r_j - r_i) \leq n\beta S; \quad \pm(r_i - r_j) \leq \frac{n\beta\bar{S}}{n-1}. \tag{A30}$$

The fixed form is the projection of a common min/max ratio band. The allocated extrema columns do not receive that extended formulation in the shared path. All increasing pairs, forward then reverse, are generated. The ideal implication chain is (A29)+absolute-value lower rows → variable-S → fixed-S. Redundancy in ideal arithmetic is not an ablation and authorizes neither row deletion nor an individual speedup claim. The cap/floor, H-spread and signed-difference rows constrain different auxiliary completions even though they preserve the same tight physical embedding.

A.5.6 Required movement and penalty closure (C1-C3)

For each station, emit only the applicable bound consequence:

$$\sum_k (d_{ki} - p_{ki}) \geq l_i - b_i \quad (l_i > b_i), \qquad \sum_k (p_{ki} - d_{ki}) \geq b_i - u_i \quad (u_i < b_i). \tag{A31}$$

These follow by substituting inventory balance in $Y_i \geq l_i$ or $Y_i \leq u_i$. Their scope inherits the domain: a cutoff-derived l_i is not an unrestricted physical bound. If $b_i \notin \mathcal{Y}_i$, A/B forces service, consistently with required movement. If $\mathcal{Y}_i = \{b_i\}$, A/B forces no service. A singleton zero domain and zero operation bounds require no divisions.

For all active finite positive-cutoff requests write

$$g + \lambda P \leq c, \qquad P \geq \underline{P}, \qquad e_i \leq \max(0, B/\omega_i)\ (\omega_i > 10^{-12}). \tag{A32}$$

The last row uses the same budget activation tests as A19. The floor follows station by station from the distance to 1 in (A21); the original e upper bound is retained and the budget row intersects it. $\alpha + \lambda\underline{P} > c$ proves ideal cutoff infeasibility. The frozen implementation instead tested $\alpha + \lambda\underline{P} \geq c - 10^{-9}$ and wrote an impossible row. Equality to c is not infeasibility of the non-strict cutoff set, so that row fails the inclusive embedding contract. The replacement deletes it in both emitters and keeps the legal floor; A.7 states the exact boundary check and the separate version qualification. The factory's "Global" metadata on the incumbent row means the entire current model, not all physical objectives at every cutoff.

Historical Round128 clarification (the ordinary rule was unchanged at that stage): if the actual cutoff-domain bound is $L_a = \alpha + \lambda\underline{P}$, its unconditional regional bound is $\min\{L_a, c\}$ by coverage with the cutoff complement. $L_a > c$ excludes the inclusive cutoff domain; $L_a = c$ permits equality and excludes only strict improvement. For $c - 10^{-9} \leq L_a < c$, the raw supported bound remains L_a and does not rule out every arbitrarily small improvement. The impossible-row shortcut is not a physical embedding inequality for all $F \leq c$. A known witness contradicting its alleged own-domain bound takes priority for investigation. The separately defined MD-R omits only this named shortcut and checks its own Start against the inclusive root; no frozen production numerical policy is upgraded by that control.

A.5.7 Scalar objective estimate and SP lifting (C4-C6)

For $S > 0$, multiplying the true cutoff inequality by nS gives

$$A + n\lambda SP \leq ncS. \tag{A33}$$

Since $A \geq 0$, $\bar{S} \geq S$ gives $A/(n\bar{S}) \leq A/(nS)$, so $A + n\bar{S}\lambda P \leq n\bar{S}c$ when $\bar{S} > 0$. Equivalently A33 implies $A \leq nS(c - \lambda P) \leq n\bar{S}(c - \lambda P)$, because the original cutoff and nonnegative G imply $c - \lambda P \geq 0$. The emitted scalar row is

$$H_\Sigma + n\bar{S}\lambda \sum_i \omega_i e_i \leq n\bar{S}c \quad (\bar{S} > 10^{-12}). \tag{A34}$$

The bound on S is the rectangular upper estimate (A21), not a new optimum of S. Tight H/e complete every physical point. If $S = 0$, then $A = 0$, $P = \sum_i \omega_i$, and the original condition is $\lambda P \leq c$. Multiplication alone makes A33 tautological and cannot enforce that condition. Thus the original objective cutoff (A32) remains essential. A34 also holds directly at S=0 if that cutoff holds.

Introduce $W = W_{SP}$ and take $W = SP$ on physical completions. The four nonnegative products on the rectangle are $(S - \underline{S})(P - \underline{P})$, $(\bar{S} - S)(\bar{P} - P)$, $(\bar{S} - S)(P - \underline{P})$, and $(S - \underline{S})(\bar{P} - P)$. Expanding them proves respectively

$$
\begin{aligned}
W &\geq \underline{S}P + \underline{P}S - \underline{SP}, \\
W &\geq \bar{S}P + \bar{P}S - \bar{S}\bar{P}, \\
W &\leq \bar{S}P + \underline{P}S - \bar{S}\underline{P}, \\
W &\leq \underline{S}P + \bar{P}S - \underline{S}\bar{P}.
\end{aligned}
\tag{A17}
$$

These are the standard McCormick envelopes (McCormick 1976), derived here without attributing an unread theorem number. They are relaxations, not an exact continuous-product identity. The writer bounds W by $w_l = \max(0, \underline{SP})$, $w_u = \max(w_l, \bar{S}\bar{P})$, and emits

$$
H_\Sigma + n\lambda W - nc \sum_i r_i \leq 0. \tag{A35}
$$

Every coefficient is explicit: H coefficients 1; r coefficients $-nc$; W coefficient $n\lambda$; e coefficients in each McCormick row are the corresponding S coefficient times ω_i. The active objective remains $g + \lambda \sum_i \omega_i e_i$ with **ObjCon=0**; W is absent from the objective. Equal S or P bounds cause valid degenerate envelopes. Zero S forces the physical W to zero but does not remove the original penalty. These rows preserve a joint tight completion of g,e,H,W and must not be claimed to preserve all loose auxiliary assignments. No GS or disaggregated-SP lift is enabled.

A.5.8 Symmetry, skipped rows and dependency order

With the active symmetry option, if **all** capacities are equal and $m > 1$, the writer adds $\sum_i z_{ki} \geq \sum_i z_{k+1,i}$ for adjacent vehicles. Common T, travel and handling make those vehicles interchangeable; relabel routes by nonincreasing visits to obtain an objective-preserving representative. Heterogeneous capacities skip the whole writer family. Start normalization may sort within equal-capacity classes, which is harmless relabelling but does not add mixed-fleet symmetry rows. A physically feasible labelled route need not itself satisfy the symmetry rows before relabelling.

Bit-only interval envelopes are $\alpha b \leq w \leq \beta b$, $g - \beta(1-b) \leq w \leq g - \alpha(1-b)$. They follow by the cases b=0,1; they are **not emitted in active VD-P**, whose q block already embeds its interval bounds. Dormant receiver-source covers, Benders, pricing, D1, exhaustive subset duration, dynamic resource cuts, custom branch priorities, S buckets, GS, AMF/RC, and old timed quanta are not part of this specification. Monitoring/target callbacks do not turn the static strengthening into dynamic separation.

The proof dependency is: physical core → implicit integer service and conservation → reachability/transfer/support; budget plus reachability → rectangle → movement/penalty/S bounds; triangle lemma plus true-G membership → Gini rows; rectangle plus original cutoff → SP/scalar rows. A/B use unique nonzero one-way service independently of the implicit-integer lemma. All can be satisfied by the same physical completion, which avoids circularly assuming the validity of a tightened domain to prove its own construction. Analytic validity and actual partial support selection are separate from computational contribution; no family-level performance attribution is made without an ablation.

A.6 Cutoff and coverage proof

Lemma A3 (cutoff complement). Suppose a returned lower bound ℓ is qualified for a model containing objective-preserving embeddings of every physical point with $G_{\text{true}} \in I$ and $F \leq c$. Then $\min(\ell, c)$ is an unconditional regional lower bound.

Proof. A point in the cutoff set has $F = f \geq \ell$. A point outside has $F > c$. The minimum covers both. If the cutoff model is infeasible, only that cutoff set is empty, so c, rather than $+\infty$, is the original-region floor. □

Lemma A4 (complete cover). Let every possible improving true-G value belong to at least one registered unit I, with a qualified unconditional floor β_I. Include the analytic complement, whose applicable floor is U. Then

$$L = \min\{U, \min_{I \in \mathcal{C}} \beta_I\} \tag{A18}$$

is a lower bound on the original optimum.

Proof. Any feasible point is in the complement or at least one covering region. Its corresponding floor bounds its objective. Taking the minimum is safe for the union. Initially $F \geq G_{\text{true}}$ allows $[0, \min((n-1)/n, U)]$ as the improving Gini range. Replacing $[\alpha, \beta]$ by both closed children $[\alpha, \mu]$, $[\mu, \beta]$ preserves coverage, including the shared endpoint. Replacement commits both children atomically with their evidence. An unsplit parent, unresolved leaf, or qualified closed leaf remains represented. Induction over updates proves the invariant. □

Bounds on the same physical domain can be combined by maximum; the union of sibling domains requires minimum. Transfer to a narrower domain requires containing scope and a handled cutoff complement, and does not construct that domain's feasible Start. Proof identities include actual model SHA, input, call, epoch, domain, cutoff and return. A local terminal status cannot certify the whole original problem. Depth/width/scheduling limits can select a request or leave an obligation open; they cannot delete it.

For a compatible own fleet, the mapped complete Start includes unused vehicles and all auxiliaries. Setting $g = G_{\text{true}}$ outside a leaf is not a valid exact-g Start. Submission, API acceptance and an improved native incumbent are different observations. No optimality proof follows from H, repeated physical U or Start acceptance.

A.7 Numerical implementation boundary

Lemmas A1-A4 concern ideal exact arithmetic and valid mathematical domains. Stored matrices use binary64 arithmetic and Gurobi 13.0.2. Native FeasibilityTol and OptimalityTol are 10^{-6}, IntFeasTol 10^{-5}, requested MIPGap/MIPGapAbs zero; retained evidence closure uses 10^{-7}. Ordinary native/outer certificates are numerical certificates under this policy. The stock-based independent zero theorem and the rational tiny-reference proof have their separately stated exact premises.

A historical inclusive-cutoff writer could serialize an artificial contradictory row at a legal cutoff boundary. R133 removed that row while retaining valid penalty and domain rows. Its exact boundary fixture and finite qualification remain in the supplement; historical raw and performance denominators are not rewritten. The final implementation retains the corrected writer. For a native-derived bound or closure, the complete physical embedding, applicable cutoff complement and qualified source are required even when an analytic floor independently exists.

Every final selected own fleet is physically verified. Ordinary positive numerical certification additionally requires qualified source/return chains and complete coverage, normal lifecycle completion and a signed gap in $[-10^{-7}, 10^{-7}]$. Rejected native proof dependencies cannot close a domain. A sufficient independent original-objective zero proof can instead establish optimality while the native-source flag stays rejected. A near-zero positive objective is not exact zero. The finalizer, normal process completion and required output belong to the production clock; later postproduction checks cannot backdate a certificate.

The source-rejected zero observations in R132/R133 keep their historical unqualified certificate/performance decisions. They are distinct from the newly executed final-panel observations under the independent-proof rule. Historical numerical anomalies and readout repairs are collected in the supporting material, with unavailable hidden-tree causes left unresolved.

The total process deadline makes a run finite, rather than guaranteeing certification in every finite window. Finite physical space and ideal complete search provide no linear forecast from a remaining objective gap.

A.8 Writer and variable map

Appendix group	Frozen source authority / mapping
Data, travel, physical objective	Parser.cpp; Evaluator.cpp; Result.cpp physical objective evaluator around line 109
Variables and VTypes	CplexBaseline.cpp 890-1005; $a \leftrightarrow mName$, $Y \leftrightarrow yName$, $\ell \leftrightarrow lName$, $H \leftrightarrow hName$, $Z \leftrightarrow zprodName$
A1-A4	Same writer 1600-1695 and 1807-1824; ConnectivityFlow.cpp downstream-count embedding and Round20Current bounds
A5-A9	Same writer 1826-1932; no Round66 arc-load replacement or Round92 handling activation
A10-A11	Same writer 2241-2274 and 2329-2335
A13-A16	Same writer 2975-3154; stateName and stateGName give s, q; bitName and prodName give plain bits/products
F0 configuration and domains	PaperK1AmSf.cpp 38-114; IntervalRowFactory.cpp 391-807; writer 1697-1750, 2049-2108, 2203-2238, 2311-2327
Factory global/local metadata	IntervalRowFactory.cpp 402-451; verified-incumbent “Global” remains cutoff-scoped
M-B mode isolation	include/Round98StateService.hpp; no projected direction, linked direction, ENS-Q or other opt-in mode
AM, targets, cover and lifecycle	PaperExternalGiniTree.cpp; GurobiBaseline.cpp; ProcessPhaseLedger.cpp; MipStartMapping.cpp

The linked implementation snapshot is the historical source authority for this unchanged formulation map. The final implementation and its exact byte capsule are identified by candidate_manifest.json and source_publication.json in the current reproduction entry. Line locations aid review; they do not replace derivations. This appendix supplies constraint groups and validity premises without claiming every engineering identifier as a mathematical contribution.

Appendix B. Executable method specification

Current frozen method. S2/MS2 uses the unchanged H and controller specified below with uniform pair-only support. Historical full-M-B traces and development variants retain their named identity. S2R is a diagnostic deletion control, not a second proposed method.

This supplement defines the controller and finite feasible startup H shared by the adopted S2/MS2 method and its full-M-B predecessor. The current S2 row policy is uniformly pair-only; the underlying H and controller definitions are unchanged. Algorithm 1 appears in Section 3; its two complete subprocedures are given below in B.0; this appendix supplies their data, numerical decisions, decoding and return semantics. Mathematical symbols refer to Appendix A. Historical names are source-map aliases, not additional algorithm variants. The definitions and proofs are retained from R125. The historical Round126 panel used the fixed full-M-B predecessor and cold P-GRB; it is retained under its own identity in the R130 historical account.

B.0 Complete leaf and cover subprocedures

```
Algorithm 2  ProcessLeaf / ChooseAction
Precondition: selected unit is still represented; no pending native call.
1  Validate/rebuild artifact for current epoch; reuse a qualified same-epoch
   parent LP or request its complete LP relaxation.
   On deadline/invalid LP return OPEN_STOP/ERROR, with parent unchanged.
   On qualified infeasible LP return CLOSE(parent,cutoff evidence).
   Merge a finite optimal parent-LP bound on this same domain.
   If b >= current U-epsilon return FATHOM with its bound source.
2  Apply NEXT rule (M5) in its stated priority order:
       noncontrolling -> REQUEUE;
       strict target -> Native(parent MIP,target NEXT), then RETURN_NATIVE;
       otherwise proceed to 3.
3  If depth>=8 or width<=1e-4+1e-12:
       Native(complete parent MIP,no target), then RETURN_NATIVE.
4  Form midpoint children [alpha,mid],[mid,beta], ids parent.0,parent.1.
   Obtain both complete current LP results or reuse B.2's valid cached pair.
   A child starts with max(parent bound,its qualified finite LP bound),
       but retain raw child LP fields for AM. Children are not active leaves.
   On either incomplete/invalid result return OPEN_STOP/ERROR; keep parent.
5  If either child is qualified infeasible: return SPLIT(children,empty flags).
   Otherwise compute score (B2) in B.3, b+=min(raw child bounds):
       b+<=b+epsilon -> Native(complete parent MIP,no target);
       score+score_tolerance>=.08 -> return SPLIT(both children);
       otherwise -> Native(parent MIP,target b+,kind CHILD).
   For a native action return its actually finalized outcome and source identity.
Postcondition: no unresolved domain removed; no target value admitted as proof
   merely because it was requested; no speculative child counted as active.
```

```
Algorithm 3  FinalizeAndUpdateCover
Input: ProcessLeaf outcome, current cover, own witness/epoch/runtime caches
1  For native outcomes require actual return and request cleanup first.
   On failed engineering qualification: stop ERROR; leave unresolved region.
   Admit any finite qualified native bound only on its recorded domain/cutoff.
   Retain the source cutoff for proof accounting in A.6; do not clip the
       raw stored numerical bound (B.4).
2  Replay a candidate fleet and original objective. If own U improves by >1e-9,
       replace own witness, increment epoch and tighten current cutoffs.
       Keep qualified old scoped floors; invalidate retained models/LP/children
       on next access using B.2; never transfer a child-only floor upward.
3  SPLIT: validate parent identity, both child ids/lineage/endpoints, shared
       midpoint, inherited bounds and full union. Install both in one transaction,
       mark parent Replaced, then mark proved infeasible children Empty.
       Discard replaced/empty backend models; failed transaction retains parent.
4  CLOSE/FATHOM or qualified native OPTIMAL/INFEASIBLE: close only this
       cutoff obligation and keep its complement floor and evidence.
   Target reached + termination requested + bound>=target-epsilon:
       set NEXT milestone only for NEXT, then requeue parent (no forced split).
   REQUEUE: retain all state/coverage, return to selection.
   TimeLimit/interrupted before proof/target: retain eligible partial bound,
       keep unit Open, stop common lifecycle. Unknown/error is never Empty.
5  Recompute full-cover minimum; output unresolved units if any remain.
Postcondition: every potential physical improver is still covered by an open
   obligation or a qualified closed/complement floor.
```

B.1 Active entry and parameters

The inherited settings below originate in the saved R123 full-M-B command. The current S2 command additionally selects `--round127-mode ms2`, so every three-station support predicate is false. Both versions select `gcap-frontier`, preset `research-round83-vds-equal-net-exchange`, `round98-state-service=m-binary`, `round90-lp-g-split=false`, candidate mode off, witness audit on and H verified-zero stop on. The preset recursively selects joint-seeded interroute decoded descent, VD-P and verified Start; the canonical paper configuration supplies the static row package and first-class interval controller. R122 uses this same M-B method with the panel's declared limits/Seeds. Input, lambda, capacities and physical times remain data, not tuned choices in this round.

Item	Frozen active value / interpretation
Initial geometry	One true-G interval $[0, \min(U, (n-1)/n)]$, n>1; n=1 has G=0
Split	Midpoint; two closed children; depth <8 and width $>10^{-4} + 10^{-12}$
Score	Balanced normalized closure, threshold .08; exact formula B.3
Infeasible child policy	Two-child split, then mark proved children empty; contraction_ enabled=false
Native work	Gurobi 13.0.2; one thread; Seed from arm; PresolveAuto; requested MIP gaps 0
Numerical policy	Scheduler epsilon 10^{-7}; FeasibilityTol/OptimalityTol 10^{-6}, IntFeasTol 10^{-5}
Cutoff	Current own U; writer epsilon=0; ObjCon=0
Resource	One process deadline; 30 s shutdown reserve in the saved runs; preparation and H are charged
Products	VD-P; continuous p/d; binary direction retained; A/B on; integer loads retained
Row generation	Uniform two-station support, no three-station support; static physical plus shared interval factory; no dynamic user/lazy cut separation; native Gurobi cuts remain available

Item	Frozen active value / interpretation
Domain option vs execution	Requested iterative/2; operative one budget intersection plus reachability intersection, recomputed by shared factory; no feedback iteration
H orders	24 shuffled chromosomes from mt19937 seed 20260626; appended constructive order; no genetic generations
H decoder	Full compaction mode 1; cache limit 200000 complete decoded orders; no generation/stagnation quota in this active branch

The "interval-mip-core-no-exhaustive-subset-duration" policy is overridden by the VD-P preset `round55-vd-p`, still excluding exhaustive subset duration. Compiler identity is the frozen MSYS2 UCRT64 g++ binary SHA256 `8cd0aa578687e4182532222c8734439b3f34ac78b6ce23cba1071c5838025fc1`. Library shuffle and equivalent-key sort order are implementation-version details; the paper does not invent a canonical cross-library tie rule absent in the source.

The accepted S2 reference and its P/PS controls are fixed in the R134 candidate manifest, confirmation commands and weight commands, together with the inherited R130 configuration. R134 uses $\lambda = .15$ in confirmation and $.075, .30$ in its added weight positions. R135's candidate manifest records the same method recompiled with a default-off AP flag; its first-stage and conditional-stage commands contain only S2/AP at $\lambda = .15$. Every AP command appends –round135-aggregate-penalty to the S2 vector below. INPUT, OUT, physical T, cap B, B_MINUS_6 and SEED take their registered values. H seed 20260626 and native settings remain fixed. The zero-stop flag does not assert return before the native lifecycle: an own zero U can produce the degenerate [0,0] root, and the independent proof is finalized after resource release and complete physical verification. The supervisor starts the production clock before admission and seals required output; native Runtime alone is not T_cert.

```
PE --input INPUT --lambda LAMBDA --T T
   --pickup-time 60 --drop-time 60 --time-limit B_MINUS_6
   --process-wall-time-limit B --process-shutdown-margin 30
   --threads 1 --mip-threads 1 --gurobi-seed SEED --gurobi-presolve -1
   --method gcap-frontier
   --algorithm-preset research-round83-vds-equal-net-exchange
   --round90-lp-g-split false --round61-candidate-mode off
   --round65-witness-audit true --round65-hga-zero-stop true
   --out OUT/result.json --log OUT/native.log
   --process-phase-ledger OUT/phases.csv
   --external-gini-artifact-dir OUT/external
   --primal-heuristic-generation-log OUT/hga.csv
   --progress-log OUT/progress.csv --native-evidence-dir OUT/journal
   --round98-state-service m-binary --round127-mode ms2
```

B.2 State, identity and order

A coverage unit has id, interval endpoints, parent id, child index, depth, base/inherited bound, current bound and sources, cutoff, status, and replacement flag. Active statuses are Open (Invalid may remain selectable as an unresolved obligation); Closed, Empty and Fathomed retain their proof role. Replaced parents are represented by their committed children. TerminalReady/Coalesced are dormant sibling-coalescing states, not normal M-B actions. Native status is not interchangeable with coverage status.

Runtime state separately contains canonical artifact path/SHA/row signature, generation and incumbent epoch; LP result (terminal-valid, optimal/infeasible, bound available, raw bound), LP epoch/SHA/endpoints; cached speculative child objects; terminal-started state; native phase count; and the NEXT milestone bit. Own witness state contains complete routes, integer operations, inventory, recomputed U and true G. The model's epigraph G

never replaces the latter. Epoch starts at zero and increments on an independently verified incumbent improvement exceeding 10^{-9}. Callback observations and physical witnesses retain separate identities.

Selection uses the **current own U** as cutoff, not an old source cutoff. It considers unreplaced Open/Invalid units with α<U−epsilon and b<U−epsilon. Find their minimum bound and form the epsilon plateau $b_I \leq b_{min} + \epsilon$. Sort the plateau by decreasing width, increasing lower endpoint, increasing upper endpoint, then lexicographic id (the first three comparisons use 10^{-12}). Cycle through this order. Reset the cursor when the plateau membership/order changes or its minimum changes by more than epsilon; after a full cycle increment the tie round. Effort, elapsed time and requested_quantum fields do not affect this `selectNextByBoundOnly` call.

On a new epoch, before reusing a selected artifact verify its old saved bytes, discard its retained backend model, clear artifact/LP state, terminal flags and child cache, and build a new generation at the current cutoff. Qualified old-domain lower bounds remain on their covering units: a smaller cutoff does not invalidate a bound already completed by its old cutoff complement. The NEXT milestone bit is initialized false in each newly created runtime and set true only by a returned NEXT target. **It is not reset in the active epoch invalidation block.** Child-target returns do not set it. No stronger child bound is sent upward to the parent merely because the child has a higher value.

Parent LP reuse requires complete evidence and the same incumbent epoch. A retained artifact must exist and match its SHA. The ordinary cached-child reuse gate requires the parent's ready flag, exactly two cached children and complete child runtime LPs; epoch invalidation clears that flag before reuse. The more elaborate endpoint/path/SHA `validRound90LpGCachedChild` check is conditional on round90 and is off here; its existence is not claimed as an active gate. When reusing children, their stored bounds inherit max(old child bound,current parent bound) and their cutoff is refreshed. AM still reads the **raw LP bound fields**, not those inherited maxima. Missing/incomplete cache falls back to constructing and solving both children; it is never interpreted as infeasible.

B.3 NEXT, child target and terminal decisions

Call NEXT after the selected parent has completed its LP and merged any qualified bound; re-read the current bound b. The function first rejects nonfinite bounds, and first requeues if any other relevant bound satisfies $b_j + \epsilon < b$, even if the milestone was already reached. Otherwise, if the milestone is false and any $b_j > b + \epsilon$, return NEXT with $t = \min\{b_j : b_j > b + \epsilon\}$. If there is no such bound, or the milestone is true, allow lookahead. The caller supplies only other **Open/Invalid schedulable units** whose current bound is below current U−epsilon (and whose interval remains relevant). Closed, Empty, Fathomed and replaced units are excluded from NEXT competitors even though their floors remain in proof coverage. Thus a lone closed unit at .40 does not supply a NEXT target to an open unit at .20. Source-cutoff metadata governs bound scope, never this current scheduling filter. The function does not use a timed quantum.

If geometry is ineligible, proceed to terminal MIP after NEXT permits lookahead. If eligible, form closed midpoint children in index order 0,1. Each is speculative until the complete replacement transaction. Solve both complete LP relaxations, or use the current admissible cache. A terminal valid child means either proved infeasible, or optimal with an available finite bound; TimeLimit/UNKNOWN is not terminal valid. Deadline exits retain the parent; an unexpected nonterminal/invalid LP is an engineering failure with the parent retained, not permission to prune it. The optional UNKNOWN→core-due research branch is off.

If either qualified child LP is infeasible, AM immediately requests a two-child atomic split. With contraction=false this applies also when both are infeasible; only after installing both does the controller mark the infeasible child or children Empty and discard those backend models. The inactive one-child contraction/both-child parent-close code is not part of M-B.

For two finite optimal LPs with raw bounds $\ell 0$,$\ell 1$, use

$$\Delta = \max(U-b, \max(\epsilon, 10^{-12})), \quad v_j = \min(1, \max(0, (\ell_j - b)/\Delta)), \quad \eta = \min(v_0, v_1), \quad \mu = (v_0+v_1)/2. \tag{B1}$$

$$\begin{aligned} s &= \max(1, |b|, |\ell_0|, |\ell_1|, |U|), \\ \delta_b &= \max(\epsilon, 32\epsilon_{mach} s), \\ \delta_{score} &= \min(1, \delta_b/\Delta + 32\epsilon_{mach}), \quad \epsilon_{mach} = 2^{-52}. \end{aligned} \tag{B2}$$

Put $b_+ = \min(\ell_0, \ell_1)$. If $b_+ \le b+\epsilon$, return terminal; else if $\eta\mu+\delta_{score} \ge .08$, split; else request a **parent-domain MIP** with bound target b_+. A child target is an intended stopping threshold, not a bound to merge into the parent before native evidence returns. A returned target requeues the parent and re-evaluates the current rule; it does not force a delayed split. This distinguishes NEXT (competitors), child target (lookahead), and terminal (complete parent MIP without a bound-target stop). The recorded adaptive rho is $\min(1, .08/\max(\mu, \max(\epsilon, 10^{-12})))$; it is diagnostic, not another active score rule.

B.4 Native requests, Starts and returned actions

An LP request relaxes the actual original VTypes, keeps the complete row set, and retains the model when permitted. A MIP request restores the exact saved VTypes before optimization. Targets and terminal requests receive the remaining common deadline after preparation, with native TimeLimit set/read back; no new per-leaf budget is created. Native bound callbacks can request termination when the qualified target is reached. Such a request is not the native return itself.

For enabled verified Start, normalize the current own fleet only among equal-capacity vehicle classes, stable sorting by decreasing number of operations and assigning smaller vehicle ids first. Clear explicit previous Starts even if the replacement is ineligible. Reverify routes, true G and F. Check interval membership with 10^{-8} and non-strict cutoff with $10^{-8}\max(1, |F|, |c|)$. An outside-interval/cutoff fleet is simply ineligible, and the MIP can run without it. Other mapping failures fail the request's engineering gate; they do not establish infeasibility.

For each used route set selected arcs to one, visit z=1, direction to pickup indicator, quantities to its integers, load to each prefix and rank to position. Flow along a q-station route is q,q−1,…,1,0. All absent route columns, including unused vehicles, are zero. Set (Y,r,e,H,G,Z) to physical values, state selector to one at Y, state_g to true G there and zero elsewhere, W_SP=SP, and the unused extrema to actual min/max ratios. Check every actual column bound/type, every actual linear row at 10^{-7}, and actual objective including ObjCon at 10^{-7}; require minimization and no unexpected nonlinear structure. Then submit a complete Start and read it back. Submission, native-log acceptance, integer-vector observation and a later improved incumbent are different observations.

The outcome object is consumed only after the optimize call returns and request finalization detaches callbacks/settles native data. Retaining the same in-memory model can preserve useful solver state, but this paper does not promise that the entire native branch-and-bound tree survives type changes or later calls. Once returned, merge finite qualified bounds only on the actual interval/cutoff; replay a proposed fleet before accepting an improved own U. This order precedes the next controller transaction.

Returned condition after required identity, finalization, zero-gap-setting and consistency gates	Coverage action
LP optimal with finite qualified bound	Merge into that parent, or save on speculative child; child alone changes no active cover
LP infeasible	Parent Empty, or speculative infeasibility retained until atomic split
Target MIP OPTIMAL with bound / INFEASIBLE	Close selected cutoff obligation; retain cutoff-complement floor; discard its model and speculative child models

Returned condition after required identity, finalization, zero-gap-setting and consistency gates	Coverage action
Target reached flag, termination-request flag, available bound ≥target−epsilon	Merge returned bound; set milestone only for NEXT; requeue; preserve parent and siblings
MIP interrupted/TimeLimit before target	Keep qualified partial bound if engineering gates passed; leave region open and end the common lifecycle
Terminal MIP OPTIMAL / INFEASIBLE	Close its own cutoff obligation; never certify siblings by this status
Terminal MIP interrupted	Leave region open and stop
Unsupported status, missing optimal bound, failed identity, incomplete LP or native error	Stop with explicit failure; preserve unresolved coverage; dependent unqualified evidence is not admitted
No launch because deadline expired	Leave region and previous evidence unchanged; finalize open result

The frozen implementation merges same-unit bounds with max(old bound,new bound), without clipping to U. Such monotonic accumulation does not itself support revocation of a later-refuted source; replacement qualification must audit actual dependent state rather than simply rename the trace. Empty/Fathomed stores max(previous bound,cutoff); Closed keeps its qualified returned floor. The raw returned L is the scheduler's complete-cover minimum, also **without clipping to U**. Tiny negative signed gaps can therefore occur and remain recorded. On the empty relevant-set path the scheduler helper alone returns 0; the enclosing lifecycle/closure logic determines the outcome.

For the exact mathematical proof, each cutoff-bound source has unconditional regional floor min(bound,source cutoff), and the analytic complement contributes its cutoff floor (A.6). This describes proof accounting, rather than an extra min-assignment performed by the frozen scheduler. Raw numerical evidence and exact proof-accounting expressions must not be interchanged. The final implementation retains the R133 source qualification and corrected inclusive-domain writer, as explained in A.7 and B.4.1. Its independent zero proof is separate from native-dependent coverage; the supplement retains the historical correction evidence. Remaining numerical limitations still apply.

B.4.1 Final independent and native proof rules

Every own final fleet is independently checked against the original physical problem. An exact stock proof of equal ratios and zero positive-weight target penalties establishes F=0; the shared production predicate additionally repeats complete per-vehicle physical verification. It does not accept a merely near-zero positive objective. P, PS and S2 use the same predicate. P obtains no H, while PS and S2 pay their complete own startup.

An independently valid full-domain nonnegative floor may prove a zero fleet even after native-source rejection. The rejected native journal and conservative restoration of the full unresolved domain remain unchanged; this is not a successful native proof or a repaired hidden search tree. Without sufficient independent proof, source-rejected positive outcomes retain the restored open domain. Ordinary qualified positive native and outer coverage certificates retain their original policy.

The proof is finalized after the final selected own fleet is known, necessary physical checks have finished and native resources are released. Required outputs and normal process completion are included in the complete production clock. Postproduction verification has its own elapsed time and cannot backfill a proof or an earlier production timestamp. Engineering source anomalies, the finite R133 checks and historical bridge tables are collected in the supplement.

B.5 Teaching states and a separately sourced event

These first numbers are **synthetic teaching states**, not measurements. Take U=1, epsilon=10^{-7}. With selected b=.20 and competitor bounds (.20,.35,.60), the plateau contains the tied .20 units and NEXT=.35. After a qualified NEXT return b=.35 the milestone is true; if all other relevant bounds are at least .35, lookahead is permitted. (If the tied .20 unit still exists, the requeue check wins first.)

For a fresh controlling parent b=.20 with no strict competitor, children (.21,.30) give Δ=.8, v=(.0125,.125), score=.000859375, b+=.21: run parent MIP to .21. Children (.20,.90) instead give b+=b: terminal, regardless of the strong sibling. Children (.50,.60) give v=(.375,.5), score=.1640625: split, install both children, with union floor .50. If the other covering unit has floor .25, global L remains .25. A target TimeLimit at .205 keeps the parent open at its qualified .205 bound; a missing Start changes none of these implications.

The following is an **actual R123 saved event** from M-B's `adaptive_mass_decision_ledger.csv` and `paper_tree_events.csv`, selected as a finite trace excerpt. On L0.0.0=[0,.09308544387476185], depth 2, b=.33371261877766062, raw child LPs (.34112380854511071,.35882784229299913), U=.3723417754990474 give eta=.19185481632186216, mu=.42100858578645295, score=.08077252489598688 and score tolerance=.0000025887181797865216. The decision is split; the event at telemetry 127.2749697 records atomic replacement and global floor .34112380854511071. Earlier recorded splits at 38.3552841 and 72.829994 used the infeasible-child two-child branch. These times are native-run telemetry, not newly measured clocks or a reconstructed unknown checkpoint. Compact source excerpts and hashes accompany the finite checks.

B.6 H encoding and top-level procedure

Physical routes are labelled lists $0, i_1, \ldots, i_q, 0$ with one signed integer operation $a_i = p_i - d_i \neq 0$ per station; missing vehicle means empty route. H's **order encoding** is different: M ordered lists partition all n stations, including a tail that the decoder may not execute. A chromosome is a permutation of station ids 1..n and M−1 identical separator tokens; splitting at separators preserves empty vehicle lists. Reordering an encoded tail does not itself describe a feasible service plan. Decoding returns signed operations; materialization deletes zero-operation stations, closes used routes at depot and verifies the entire fleet.

```
H(instance, common deadline):
    C := empty physical fleet, verify Y=b
    repeat choose and admit best joint insertion (B.7) until none/deadline
    keep verified C as an own candidate
    generate all 24 shuffled order chromosomes with mt19937(20260626)
    append C's order after deterministic tail completion (below)
    initialize a shared complete-decode cache, empty
    decode all 25 orders in order, publishing verified improvements
    for each seed in order:
        repeat generate guided candidates (B.10), order by proxy fitness
        fully decode candidates in that order
        accept first fitness improvement >1e-12; restart this seed's pass
        if every candidate checked without improvement: finish this seed
        honor verified-zero/deadline between calls and candidates
    take the best independently verified candidate, including C
    run strict insertion/quantity closure followed by best neutral block move;
        repeat these two phases (B.11) until zero/no neutral/deadline/failure
    return best admitted complete fleet and stop reason
```

Constructive tail completion starts with C's visited order and each vehicle's complete physical duration as a proxy. For each unvisited station i in increasing id, append it to the vehicle minimizing proxy[k]+tau(last,i)+tau(i,0)−tau(last,0), choosing the first vehicle on an exact tie; update that proxy. The remaining operations are decided by the same full decoder. Twenty-four shuffles are generated before

appending this extra seed, preserving random draws. `constructive_mode=0` means there are no additional biased population members. The snapshot's C++ shuffle is authoritative for exact seed orders; the algorithm specifies the generator/seed/order and permits cross-library sequence differences. HGA crossover, mutation, education generations, 2000-generation stagnation and obsolete "10 iterations" fields do not run in this decoded-descent branch.

The outer physical candidate store keeps a candidate only after full verification and an improvement greater than 10^{-10}, unless no incumbent exists. Verified-zero can terminate startup early. This publication threshold differs from inner 10^{-12} move thresholds; an internally accepted tiny improvement need not replace the outer witness.

B.7 Joint insertion

At a verified fleet let $L_k(a)$ be load on leg a, numbering the original legs from 0 through the number of visited stations. For each ordered (pickup station i or 0,drop station j or 0), excluding (0,0), i=j, and already visited nonzero stations, enumerate vehicles and positions. Zero denotes an absent side, not depot stock. Both sides use the same integer quantity q. For a pair choose original legs a≤b, inserting i then j; on one leg the sequence is predecessor,i,j,successor. Travel increment is the sum of replaced arc differences (one combined expression when a=b).

The load upper amount is $Q_k - \max_{v=a..b} L_k(v)$ for a pair; $Q_k - \max_{v=a..end} L_k(v)$ for pickup only; $\min_{v=a..end} L_k(v)$ for drop only. Intersect with available inventory at i, room at j and max vehicle capacity. Complete duration increment is travel increment plus hq when a pickup is inserted, and just travel increment for drop only (that drop replaces return unloading). The time test uses T+10^{-7}. For each amount, retain the cheapest feasible travel placement, ties (vehicle,a,b), by a suffix scan of placement amount limits. Evaluate all amounts 1..limit using F on the changed one/two inventories; accept only gain>10^{-12}.

Choose globally: nonpositive added duration before positive; then greatest gain per added duration (compare cross products, or gain for both nonpositive), greater gain, smaller added duration, and lexicographically (i,j,q,k,a,b). Apply and fully replay; require objective agreement within 10^{-10} and verified gain>10^{-12}. Every insertion adds at least one previously unvisited station, so construction has at most n accepted insertions. This finite placement/amount scan is a heuristic neighborhood, not an optimization backend.

B.8 Single and paired quantity modification

Only currently served stations participate. Enumerate a increasing, then b=0 (single) and each served b>a (pair). For every nonzero integer t set $a'_a = a_a + t$, and $a'_b = a_b - t$ if b exists. Station bounds give

$$b_a - C_a - a_a \le t \le b_a - a_a; \qquad a_b - b_b \le t \le a_b - b_b + C_b \quad (b \neq 0). \tag{B3}$$

Intersect these with every affected prefix load $0 \le L + \chi t \le Q_k$, where chi is +1 after a, −1 after b, zero before/beyond the cancelling pair. This includes cross-vehicle modifications; each route has its own capacity. Enumerate all integers in the resulting interval increasing, skip zero. Convert signed operations to pickup=max(a',0), drop=max(−a',0), remove zero stops and shortcut travel. Sign reversal is allowed. Check complete duration, including depot unloading, at 10^{-7}; evaluate F; require gain>10^{-12}. Choose lowest proposed objective, tie (a,b,t). Full replay and objective agreement within 10^{-10} precede acceptance. Missing stations cannot be added by this operator; insertion handles them.

Strict closure computes both the best insertion and best quantity move on the same current fleet, chooses smaller proposed F, insertion on an exact tie, admits the verified strict improvement, and repeats. If neither exists, that union neighborhood is exhausted. Verification failure retains the previous fleet and records failure. This is exactly

the Round76 subprocedure called by active exchange descent; the standalone Round75/76 preset entry branches are not separately run by main.

B.9 Complete greedy decoder and its compensating moves

For each encoded route take its maximal prefix whose travel with direct return stays within T+10^{-12}, stopping at the first exceeding station. Fix this travel during that decoder pass. Set the cumulative pickup budget $B_k = \max(0, \lfloor (T - travel + 10^{-12})/h \rfloor)$ (an integer sentinel max-int/4 if $h \leq 10^{-12}$); this is a handling budget, not Q. Initialize all operations zero and Y=b. The single-route proxy version instead keeps other-route operations fixed. The active mode is 1: visit vehicles and stations forward, greedily match each current surplus above clipped target to later deficits, as much as remaining supply, demand, station bounds, all intervening load bounds and pickup budget allow; leftover surplus may be collected for depot return. This constructive initialization does not require each step to decrease F.

Define an interval move by vehicle k, positions a<b, sign $\sigma \in \{+1, -1\}$, amount $q \geq 1$. Increase a's signed operation by sigma q and decrease b's by sigma q. The sink may instead be the depot after the last position, in which case only a changes. Loads on [a,b) change by sigma q. Each final inventory must stay in [0,C], loads in [0,Q], and total positive operations in [0,B_k]. Thus an earlier delivery can be reduced/reversed and a pickup compensated elsewhere; there is still only one net one-way operation per physical station. Feasible amount bounds come from station room and the minimum/maximum load on the changed interval; the positive-part pickup change is piecewise linear with breakpoints at operation sign changes. The implementation finds the largest feasible q by these segments, then evaluates every positive integer up to it. The feasible amount sublevel contains q=0 and is an interval, so no integer hole is skipped. Objective arithmetic updates the changed ratios, pairwise absolute sum and penalties; it is not an LP/DP solve.

The active decoder loop is:

```
repeat:
    repeatedly apply the best strict atomic interval move, until none
    if a strict same-route composite exists: apply best; restart
    if a strict two-route composite exists: apply best; restart
    if a strict unit interval polish exists: apply best; restart
    if a strict flow-pair redistribution exists: apply best; restart
    if no atomic move occurred in this pass: return current operations
```

Atomic enumeration is vehicle, source position, later sink positions then depot, sign + then −, amount increasing. Keep the best move and the eight best atoms per route. To update the best, initialize value F−EPS and replace only when candidate+EPS<best, EPS=10^{-12}; this gives an effective initial gain test >2EPS. For each bounded list, append candidates until size K ($K = 8$ atoms or $K = 6$ local candidates), maintaining a max heap with comparator a.objective<b.objective. Once full, replace its front only if candidate.objective+EPS<front.objective, then restore the heap. Final sorting compares only ascending objective, with no secondary key. Equivalent-key heap positions consequently depend on the frozen standard library. Local candidates comprise each retained atom and each pair of distinct atoms on one route, adding their operation deltas simultaneously. Recheck all combined loads, positive-operation budget and station bounds, then retain at most six best candidates per route using the same bounded-heap objective logic. Candidates can individually worsen F: their combination may compensate that worsening. Same-route best selection uses F−EPS and candidate+EPS<best. Cross-route candidates combine one retained local candidate from each pair of vehicles, recompute the full common Gini/penalty objective for their jointly changed (up to eight) stations, and use the same strict test. Distinct routes have disjoint stations, so combined physical feasibility follows from their checked local feasibility; joint objective evaluation remains necessary.

Unit polish checks amount 1 for every legal atomic interval, with initial best F−10^{-14} and candidate+10^{-14}<best. Flow-pair polish ignores zero operations, partitions the route into consecutive same-sign runs, and pairs each

pickup run with its immediately following drop run. For each ordered pair of distinct such run pairs, choose one pickup and drop position in each: subtract one unit from the first pickup/drop flow and add one to the second. Check both interval load changes together, station bounds and original signs; pickup total is unchanged. Its candidate helper still requires gain>10^{-12}, followed by the 10^{-14} best-comparison guard. Residual pair-unit, route-exact and pair-exact neighborhoods return false in this active implementation; they are not hidden optimization oracles.

After a first full decode, if its executed prefix contains zero-operation stations, delete them, keep only its nonzero active order, and run this same full decoder from zero operations **once more** on those compacted routes. Return whichever has smaller F, first decode on exact tie. This does not repeat compaction to a fixed point and does not restore the removed tail. The cached object contains complete signed operations and fitness=−F. Materializing that object removes remaining zero operations and independently checks the original physical route duration.

Decoder cache key is compaction mode plus the ordered lists with separators; it includes tail positions, even if they decode identically. On a hit return the entire stored decode. On a miss at size≥200000 clear the whole map before inserting; this is neither LRU nor a permanent visited-state ban. Internal atomic/local caches are tied to an incremented operation-state epoch and cleared after accepted updates. The static ratio/penalty preprocessing cache keys capacities, targets and weights (including source slice offsets); it affects arithmetic reuse, not candidate admission. No general accepted-state de-duplication is claimed.

B.10 Guided order descent and ties

For each route compute the travel-only executable prefix as above. Within it record first negative-operation position and last positive-operation position. A station's role is pickup if its decoded operation>0, drop if <0; if zero, use surplus/shortage of decoded final inventory relative to target, otherwise neutral. Supply anchor is first drop, or position 0; demand anchor is immediately after last pickup, otherwise prefix end.

Generate, in vehicle order: move the last zero-operation prefix station to that route's end; collect every nonneutral prefix position plus the first supply and first demand in its tail. Move a supply lying after first drop, or in tail, to the supply anchor; move a demand before last pickup, or in tail with a last pickup present, to the demand anchor. Duplicate generated candidates are not globally removed. Same-route erase/insert decrements destination position if it originally lay after the erased position; skip unchanged order. For interroute proposals take only the first supply/demand in each source tail, each different destination in order: insert supply at destination's supply anchor if that route contains any demand-role station; insert demand at its demand anchor if it has an executed pickup.

The proxy fully runs the single-route incremental decoder with other-route operations fixed. For a two-route change try both route decoding orders and take the larger fitness (smaller F). Proxies only order proposals: sort decreasing proxy fitness, then fully decode in that order and accept the first fitness increase>10^{-12}. An unsuccessful pass evaluates every generated proposal, including a poor proxy. There is no top-k rejection here.

The guided comparator has no secondary key: equal proxy values are equivalent to C++ `std::sort`, which is not stable. Likewise internal bounded-heap/objective-only sorts have implementation-dependent equivalent-key order. For the frozen compiler/library they supply the recorded run's repeatable behavior; a cross-language implementation may choose a different equivalent-key order and consequently a different heuristic path. Specifying lexicographic ties here would change the frozen algorithm. By contrast B.7/B.8 and the physical block operators do have explicit deterministic ties. This limitation affects path-level reproduction, not feasibility, exact model scope or native lower-bound validity.

B.11 Neutral block moves and exit argument

Strict closure is followed by one best neutral move from the union below. The potential is the **complete** vector of all vehicle durations, sorted decreasingly, including zeros for unused vehicles; compare it with exact C++ lexicographic <, never by an aggregate sum or a tolerance that ignores an earlier increase.

Balanced relocation enumerates source vehicle, all nonempty contiguous half-open position blocks [a,b), other target vehicle and every insertion leg. Require net signed operation zero. If relative prefix load extrema inside the block are lo,hi, require target load+lo≥0 and target load+hi≤Q_target. Removing a net-zero block leaves source suffix loads unchanged; inserting it leaves target suffix loads unchanged. Keep operation quantities and order inside the block; recompute both full durations and accept only if each≤T+10^{-7} and the sorted vector strictly improves. Choose smallest potential, tie (source,a,b,target,leg).

Equal-net exchange enumerates every contiguous nonempty block per route, bucketed by integer net operation in increasing net order; pair vehicles s<t and blocks from equal-net buckets. At each insertion prefix require its old load plus the incoming block's relative extrema in that vehicle's [0,Q]. Equal net preserves all suffix loads, even when the common net is nonzero. Replace the two blocks in their original internal order, preserve all station operations, recompute complete durations and compare the same potential. Choose smallest potential, tie $kind, source, a, b, target, target_a, target_b$, with kind 0 relocation before kind 1 exchange. `bestRound83Neutral` evaluates the whole relocation neighborhood first and then exchanges.

Before accepting a neutral candidate, independent replay requires identical final-inventory vector, **exactly equal recomputed objective**, and exact equality between recomputed duration potential and the proposed potential, which must strictly improve the prior one. Failed admission retains the prior fleet. After an accepted neutral move, run strict closure again. Stop on exact verified objective zero, common deadline, failed verification, or closure plus no improving neutral move. A selected no-move event is an exhaustion claim only for these enumerated neighborhoods.

In ideal arithmetic, after the finite construction the accepted physical phases strictly decrease (F,sorted durations) lexicographically, and the finite space of labelled routes/integer quantities excludes cycles. The order-descent phase strictly increases its deterministic decoded fitness on a finite set of orders. The implementation uses incremental floating arithmetic and tolerances in the decoder; these comparisons are not a proof of exact rational monotonicity or an accumulated error bound. The physical block phases have exact inventory/objective equality for neutral steps, unlike a merely tolerance-neutral rule. There is no fabricated de-duplication argument. The common deadline and external lifecycle cap are the operational stopping boundary; expensive decoder calls are finite routines and do not have a separate native optimization timeout. H neighborhood closure never certifies the original optimum.

The separately evaluated inclusive-root, support and equivalent-expression controls are documented in Supplement S11 with their original evidence identity.

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